(Mis)perceptions about children

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Abstract

I quantify how early childhood teachers' perceptions of developmental delays vary with their reference group. Using standardized non-cognitive and cognitive development measures for Australian children aged 4–5, I find that, conditional on a child's own development, teachers in neighbourhoods with lower average non-cognitive development are less likely to perceive delays. The estimated neighbourhood gradient is large enough to reverse the apparent ranking of neighbourhoods. Neighbourhoods with higher measured non-cognitive development have higher reported non-cognitive delay rates, but lower predicted rates when neighbourhood-average development is held at a common value in the fitted model. Teachers' perceptions are reflected in mothers' beliefs, which in turn predict uptake of therapy. Finally, teachers with university degrees are more likely to report delays among children with low measured development in both domains, while more experienced teachers are more likely to report non-cognitive delays among children with low measured non-cognitive development.

JEL classification: I21, I24, D83, J13, J24

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1 Introduction

Correctly identifying non-cognitive (e.g., socio-emotional) and cognitive (e.g., receptive language) developmental delays in preschool-aged children is critical for educators and families when deciding how to support children's early development, whether by seeking speech therapy or psychological and behavioural support. Developmental delays occur when a child does not reach their developmental milestones within the expected time frame relative to their same-age peers. Addressing these delays in early childhood is particularly important because skills are self-productive, with early delays potentially compounding over time and hindering learning and socialization in later years (Heckman, 2006; Cunha, Heckman, and Schennach, 2010).[1] Crucially, families' biased perceptions of children's cognitive development have been shown to lead to suboptimal investment decisions, such as picking the wrong textbook level, that disrupt the accumulation of human capital (Dizon-Ross, 2019; Kinsler and Pavan, 2021; Bergman, 2021).

Two recent papers provide evidence that teachers' perceptions about cognitive skills are influenced by reference group bias, with teachers in low-achieving schools overestimating cognitive skills and those in high-achieving schools underestimating them (Kinsler and Pavan, 2021; Elder and Zhou, 2021). Both studies identify these distortions by comparing teachers' perceptions with standardized cognitive tests that provide an independent benchmark of children's skills. While non-cognitive skills are as important as cognitive skills for a range of life outcomes (Heckman, Stixrud, and Urzua, 2006; Conti, Heckman, and Pinto, 2016; Deming, 2017), at scale they are typically captured only through teachers' or parents' subjective ratings, with few independent assessments. As a result, teachers' judgments often serve as the primary source of information on non-cognitive delays for schools, governments, and even families.[2] The extent to which the local reference group shapes these judgments has, to date, only been quantified under strong as-

[1] The Centers for Disease Control and Prevention (CDC) emphasizes the importance of acting early on concerns about developmental delays to ensure that children receive appropriate support before the start of formal schooling (see CDC, "Developmental Monitoring and Screening"). Up to 50 percent of preschool behavioural problems can develop into childhood mental health problems, increasing risks of substance misuse, family violence, and crime (Luangrath and Hiscock, 2011).

[2] For example, in 2024 in Australia, over 12 percent of total government funding for primary and secondary schools was allocated to accommodations for students with disabilities, largely based on teachers' judgments. Nearly a quarter of school-aged students received accommodations, with over 88 percent addressing non-cognitive and cognitive delays. See ACARA's report and the Australian Schooling Resource Standard.

sumptions about the unobserved distribution of non-cognitive skills or the size of the reference bias itself.[3]

I address this challenge by using direct, face-to-face observations from psychologist-trained interviewers who evaluate children's non-cognitive skills in the Longitudinal Study of Australian Children (LSAC). These observations provide a structured behavioural measure of non-cognitive skills, alongside measures of cognitive skills based on language tests administered in the survey.[4] Importantly, the survey also collects teachers' private beliefs about whether a child shows signs of non-cognitive or cognitive delays relative to other children of similar age.[5] To examine whether these perceptions vary with the local environment, I follow an approach similar to that of Kinsler and Pavan (2021) and Elder and Zhou (2021), estimating a measurement system in which perceived delays are modelled as a function of both the individual development of a child and the average development level of other children in the neighbourhood.

I find that early childhood teachers' perceptions of developmental delays in children aged 4–5 are systematically related to the local environment. Conditional on a child's own measured development, a one standard deviation increase in average neighbourhood non-cognitive development is associated with a 2.1 percentage point increase in the probability that a teacher reports a non-cognitive delay, against a base rate of 20 percent, and with a 1.4 percentage point increase for perceived cognitive delays, against a base rate of 14 percent. Average neighbourhood cognitive development, by contrast, predicts only perceived cognitive delays, extending the findings of Kinsler and Pavan (2021) and Elder and Zhou (2021) from the U.S. context. These findings remain robust after a range of checks, including controls for detailed neighbourhood and teacher characteristics, which reduce concerns about the non-random sorting of households and teachers into neighbourhoods.

These magnitudes are large enough to reverse how neighbourhoods rank in reported non-

[3]Lacking an independent measure of non-cognitive skills, Elder and Zhou (2021) use between-school sorting on cognitive test scores as a proxy for sorting on the latent noncognitive skill, apply the reference bias estimated from teachers' cognitive ratings to their noncognitive ratings, or predict noncognitive skills from within-school variation, which removes reference bias by construction.

[4]This interview-based measure represents a consistently measured independent source of information about a child's development, but cannot be used on its own to formally diagnose a developmental delay.

[5]Throughout the text, I refer to all early childhood education and care (ECEC) providers as "teachers." This includes staff who, in the Australian context, are commonly classified as early childhood teachers, educators, or assistants, depending on their qualifications.

cognitive delays. In the data, teachers in neighbourhoods at the top quartile of average non-cognitive development are 1.1 percentage points more likely to report non-cognitive delays than those in bottom-quartile neighbourhoods. Holding average neighbourhood development at a common value in the fitted model, so that teachers no longer differ in their local reference group, reverses this ranking. Teachers in top-quartile neighbourhoods would instead be 4.2 percentage points less likely to report non-cognitive delays than those in bottom-quartile neighbourhoods. This has implications for governments that use nationwide teacher evaluation statistics to identify disadvantaged areas. The Australian Early Development Census, for example, collects teacher assessments nationwide to document geographic variation in early childhood development and inform community and school planning. If teachers benchmark against local peers, statistics of this kind will understate delays in lower-development areas, where they are more common, and overstate them in higher-development ones.

Having shown that teachers' perceptions vary with the local environment, I next examine whether the likelihood of reporting delays in children with low measured development also differs by teacher qualifications or by early childhood program characteristics. I find that teacher education is an important predictor of delay recognition, with teachers holding a university degree 8.5 percentage points more likely to perceive non-cognitive delays among children with low standardized measures of non-cognitive development than those with certificates or diplomas. The pattern is similar for cognitive development, with a 5.6 percentage point difference. Teacher experience, by contrast, is associated only with recognition of non-cognitive delays. Among children with low measured non-cognitive development, teachers with 0–5 years of experience are 9.1 percentage points less likely to perceive delays than more experienced teachers, while no comparable difference appears for cognitive delays.

Teacher education and experience are associated with different features of these perceptions. Perceptions of university-educated teachers correspond more closely to children's measured cognitive development, but do not vary less with average neighbourhood development. Experience, by contrast, is associated with both dimensions, but only for non-cognitive delays. Less-experienced teachers' perceptions are less aligned with children's measured non-cognitive development and more strongly related to the local non-cognitive environment.

I then examine how these perceptions relate to mothers' beliefs and to the subsequent uptake of therapy. Mothers' own perceptions of non-cognitive delays are not related to average neighbourhood non-cognitive development, so mothers do not appear to benchmark against the local non-cognitive reference group in the way teachers do. Instead, mothers' perceptions are associated with the assessments teachers and schools convey to them: mothers contacted by the school about their child's behaviour are 10.7 percentage points more likely to perceive a non-cognitive delay, conditional on their prior beliefs. These perceptions, in turn, predict the later use of professional services. Children whose teachers or mothers perceive a delay at ages 4–5 are more likely to receive therapy in the corresponding domain by ages 6–7, conditional on the child's own measured development and prior therapy use.

This paper contributes to the growing literature examining how beliefs about children's skills relate to investment decisions. Much of this work has focused on cognitive skills, where standardized assessments provide a benchmark against which beliefs can be compared and updated. In experimental settings, Dizon-Ross (2019) and Bergman (2021) show that parents hold inaccurate beliefs about their children's school performance and effort, and that providing accurate information changes both beliefs and subsequent investment. Where comparable benchmarks are unavailable, subjective evaluations play a larger role in identifying children who may need additional support. I provide evidence on how these evaluations vary with the local environment, and on how they relate to the uptake of therapy. In doing so, I also speak to why children's environments differ across parental socioeconomic status (SES) and neighbourhoods. Prior research has documented several important drivers, including resource constraints, parental preferences, and differences in perceived returns to investment.[6] I focus instead on neighbourhood-related information frictions in the evaluation of children's development.

Finally, this paper contributes to the literature on how early childhood education quality shapes children's development. While there is strong evidence that high-quality programs improve both short- and long-term outcomes, particularly for disadvantaged children, there is less consensus on which specific program or teacher characteristics are most influential.[7] Regarding

[6] See Attanasio, Cattan, and Meghir (2022) for a review of research exploring the drivers of SES gaps in children's environments.

[7] For example, while a review by Manning et al. (2019) highlights teacher education as a key correlate of program quality, Early et al. (2007) conclude that policies aimed at raising teacher education levels may not lead to quality

the identification of non-cognitive delays, Page, Gilham, and Hill (2024) find that educators' mental health literacy varies substantially by education and training. I contribute to this literature by examining the factors that predict the identification of developmental delays, an important early-stage outcome that can influence the support that children receive.

The remainder of the paper is structured as follows. Section 2 describes the data, and Sections 3 and 4 examine how teachers' and mothers' perceptions of child development vary with the local environment. Section 5 explores how these perceptions relate to therapy uptake. Section 6 concludes.

2 Data

The data for this project come from the LSAC, a national study that follows two cohorts of Australian children: the "baby" cohort (B cohort), which includes 5,107 children aged 0–1 in 2003–2004, and the "kindergarten" cohort (K cohort), composed of 4,983 children aged 4–5 in 2004. The survey started in 2004 with participating families interviewed biennially. The main analysis uses data for children aged 4–9 who attend formal care or education arrangements, with attendance close to universal at these ages.[8]

Three features of the survey allow me to investigate variation in teachers' perceptions across local environments. First, it collects standardized measures of children's non-cognitive and cognitive development during the interview, independently of teachers and parents. Second, it elicits teachers' and mothers' private beliefs about whether the same child shows a developmental delay. Third, it samples children by postcode, which allows me to characterize each child's local environment using the other children surveyed in the same postcode. The survey also provides rich information on family and neighbourhood characteristics, allowing me to control for a broad set of observable differences across families and local environments.

improvements.

[8]This sample restriction allows me to focus on the perceptions of staff with childhood education and care qualifications, who are typically employed in these formal settings. Over 95 percent of children aged 4–5 attend formal care or educational arrangements, mainly daycare, preschool, or kindergarten, as do over 97 percent of children aged 6 and above, who are within the compulsory schooling age in most jurisdictions.

2.1 Measures of child development

I use direct observations of children's non-cognitive skills recorded by interviewers during the in-person household visit to construct a standardized measure of non-cognitive development. These observations, available when children are 4–5 and 8–9 years old, are based on 1- to 2.5-hour visits during which interviewers observe the child both with and without their parent present, allowing for assessment across different interactional contexts. The interviewers evaluate the non-cognitive development of children across three dimensions: negative response (e.g., fussing, crying, vocal or physical expression of anger), focus during the cognitive test, and positive response (e.g., smiling or laughing).[9]

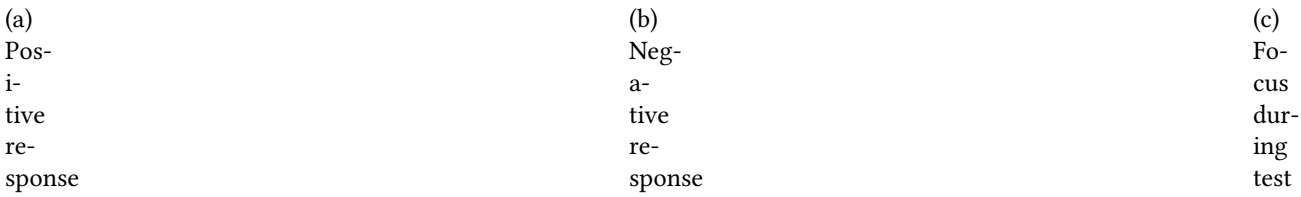
These measures have two important advantages for quantifying how the reference group relates to teachers' perceptions. First, interviewers were trained by psychologists and completed practice interviews with parents and children to ensure consistency in evaluations. Second, they used standardized rating scales that captured both the frequency and intensity of behaviours. For example, when evaluating children's negative or positive responses, they selected from five options ranging from "none displayed" to "three or more intense, heightened, or prolonged displays."[10] Structured behavioural observation methods are used in psychological assessments alongside other sources of information when diagnosing non-cognitive delays in children (Volpe et al., 2005; Minder et al., 2018). The use of a common protocol and standardized scales across all neighbourhoods helps reduce potential bias in the resulting assessments and provides a consistently defined benchmark across locations.

Figure 1 shows the distribution of interviewer-recorded observations for children aged 4–5. While positive responses varied, 63 percent of children were both constantly focused and showed

[9]Persistent loss of temper and aggressive behaviour, captured by the negative response measure, are symptoms of behaviour disorders in children (see CDC, "Behavior or conduct problems"). Similarly, difficulty concentrating and sustaining focus during tasks, as reflected in the focus during the cognitive test, is a core symptom of Attention-Deficit/Hyperactivity Disorder (ADHD) in young children, according to the Australian Psychological Society (see Australian Psychological Society, "ADHD in children"). While focus is often treated as a cognitive process in psychology, inattention is a common symptom of behavioural and neurodevelopmental issues. In the education literature, attention and task persistence are frequently used as indicators of non-cognitive skills, such as student effort and self-regulation (Rosen et al., 2010; Lundberg, 2017). In Appendix A, I show that my results are robust to measuring non-cognitive skills using only the degree of the child's negative response.

[10]The five options for the degree of children's focus include the following: constantly did not pay attention; typically did not pay attention and attended in one to two instances; did not pay attention half the time; typically paid attention but attention wandered in one to two instances; and constantly paid attention/concentrated.

Figure 1: Non-cognitive development observed during the interview at ages 4–5



Notes: The figure displays histograms of interviewer-recorded observations evaluating non-cognitive skills of children aged 4–5 attending formal care or education settings. Panel (a) shows the degree of positive response — smiling, laughing, or sounding excited, happy, or pleased. Panel (b) shows the degree of negative response — fussing, pouting, whining, crying, or expressing anger vocally or physically. Panel (c) shows the child's level of focus during the Peabody Picture Vocabulary Test (PPVT).

no negative responses during the interview. I construct a non-cognitive development score as the first principal component of three interviewer-recorded observational measures. Each measure is first age-standardized, and the resulting principal component is then re-standardized by age to ensure comparability with other skill measures used in the analysis.[11] The resulting score is left-skewed, with many children clustered at the top and a long left tail capturing those who display non-cognitive problems with varying degrees of frequency and intensity (see Figure B.1). This measure allows me to detect children with symptoms of non-cognitive delays in the left tail of the skill distribution but does not differentiate between children at the top who are more likely to be developmentally on track.

In addition to observing non-cognitive skills, interviewers assess children's cognitive development using a short form of the Peabody Picture Vocabulary Test (PPVT), a standard, age-adapted measure of receptive vocabulary and spoken word comprehension. The PPVT is commonly used in the literature as a measure of children's cognitive skills (Fiorini and Keane, 2014; Nicoletti and Tonei, 2020). Another measure of cognitive skills available in the survey is the Who Am I (WAI) assessment, which is administered for children aged 4–5 to evaluate the general cognitive abilities needed for school readiness. This assessment tests receptive and expressive language and numeric abilities, and I use it as an instrument to address measurement error in the PPVT.

A common critique of standardized tests of cognition is that they provide context-dependent

[11] See Table B.1 for the PCA loadings.

measures of children's academic abilities that depend on student effort, motivation, test-taking skills, and a range of other factors (Heckman and Kautz, 2012). The interview measure of non-cognitive skills, based on a limited set of interactions between interviewers and children, is likely to have comparable limitations, and neither measure can be used on its own to formally diagnose a developmental delay. Teachers observe children over longer periods and in more settings, and therefore have information that a single visit cannot capture. At the same time, their assessments may incorporate the children they encounter locally. The interview-based scores therefore serve here not as a superior assessment of a child's development, but as an independently collected, consistently measured one that captures meaningful variation in development and predicts future outcomes.[12]

2.2 Measures of perceived developmental delays

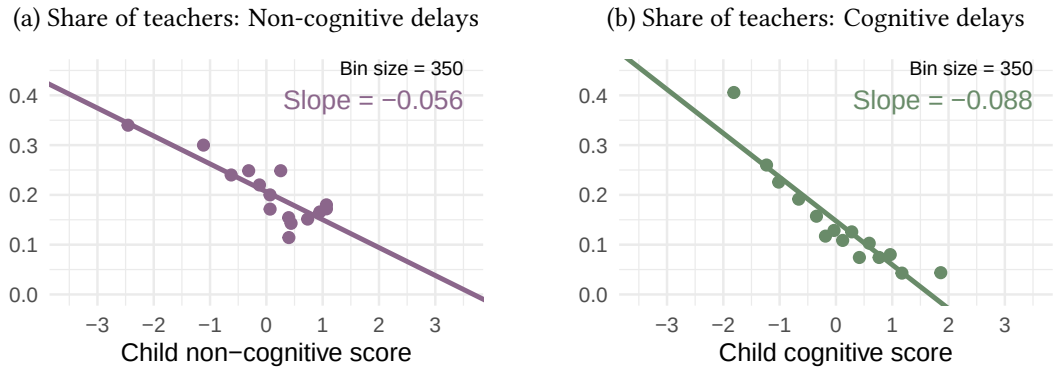
The LSAC also captures teachers' private perceptions of children's developmental delays. When a child is 4–5 years old, teachers complete a self-administered questionnaire that asks them to rate the child's development relative to others of similar age across several dimensions, including socio-emotional development (e.g., adaptability, cooperation, responsibility, self-control) and receptive language ability (e.g., understanding, interpreting, and listening).[13] These dimensions align with those assessed by interviewers, allowing a direct comparison between measured development and teachers' perceptions. The interviewers' assessments are not observed by teachers or parents.

To incorporate these teacher assessments into my analysis, I construct binary indicators of perceived developmental delays based on their responses. In the questionnaire, teachers rate the child as "much more competent than others," "as competent as other children," "less competent than others," or "much less competent than others." I define an indicator for perceived non-cognitive delay equal to one when a teacher gives one of the two lowest ratings in the domain of non-cognitive development, and define perceived cognitive delay analogously using receptive

[12]For example, children's non-cognitive and cognitive scores at ages 4–5 are associated with a lower likelihood of grade repetition by ages 12–13 and higher grade 9 national test scores in reading and numeracy (see Appendix C).

[13]The exact statement was "Rate how this child was compared with other children of a similar age, over the past few months." Teachers were asked to fill in the self-complete questionnaire and mail it in a pre-paid envelope, so parents did not observe their replies, and no teacher faced an incentive to report a more or less favourable assessment than they held.

Figure 2: Children's interview-based development scores and teachers' perceptions at ages 4–5



Notes: Panels (a) and (b) display binned scatterplots showing the share of children perceived by teachers as having non-cognitive and cognitive delays, respectively, plotted against their corresponding non-cognitive and cognitive interview development scores. Both panels report raw regression lines. The sample includes children aged 4–5 attending formal care or education settings, with non-missing interview-based non-cognitive and cognitive development scores, neighbourhood average development scores, and teacher perception measures.

language development.[14] Around 15 percent of teachers report delays in cognitive development, compared to just over 20 percent for non-cognitive delays (see Appendix Table B.2).

Figure 2 relates perceived delays to interviewer-assessed development scores. In both domains, the slope is negative: children who scored higher during the interview are less likely to be perceived as delayed by their teachers, even though teachers do not observe these assessments. Since both teacher perceptions and interview-based measures are informed by children's individual development, a similar association might be anticipated at the neighbourhood level, such that teachers in higher-development neighbourhoods would report fewer delays. Subsection 2.3 describes the construction of the average neighbourhood development score and shows that this pattern does not necessarily hold if teachers evaluate children relative to local reference points.

In addition to teacher assessments, the survey also collects mothers' perceptions of non-cognitive delays by asking whether their child is "easier than average", "about average", or "more difficult than average" compared to other children of the same age. I create a binary indicator equal to one if mothers perceive their children as more difficult compared to others of similar age.

[14]Fewer than 4 percent of teachers indicate that children are "much less competent" in either the cognitive or non-cognitive domains. Therefore, I pool the "much less competent" and "less competent" responses to indicate delays.

Only 7 percent of mothers consider their child to have a non-cognitive delay (see Appendix Table B.2). When children are 8–9 years old, mothers are also asked whether the school has contacted them about their child's behaviour within the last 12 months. Unlike teachers' questionnaire responses, this is a signal the mother directly observes, which I use in Section 4 to examine how mothers update their beliefs.

Importantly, the delay items ask teachers and mothers to compare a child's development with other children of a similar age, without specifying a class, school, or neighbourhood as the reference group. The wording therefore points to a general age-appropriate developmental standard rather than an explicitly local comparison. This interpretation is supported by the questionnaire design, which names local reference groups when they are intended.[15] The distinction is consequential, as Kinsler and Pavan (2021) show that mothers' evaluations of cognitive skills differ depending on whether they are asked to compare their child with others of similar age or with classmates. The empirical question is whether respondents nevertheless anchor their judgments on the children they encounter locally. This matters because, if they do, otherwise identical children may be assessed differently depending on where they live. It also matters because such assessments are used beyond the classroom. In Australia, teachers' subjective ratings inform national reporting on early childhood development, and similar ratings have been used as measures of child development in economic research (Frijters, Johnston, Shah, and Shields, 2009).

2.3 Measures of the local environment

A key feature of the LSAC data is the availability of household postcode information, which I use to define neighbourhoods throughout the analysis. The sampling design groups children geographically by postcode.[16] Australia has over 2,600 postal areas (Australian Bureau of Statistics, 2021), and the first-wave data include children from 409 distinct postcodes, with an average of 24.7 children per postcode. This geographic clustering allows me to construct neighbourhood-level aggregates of child development.

[15] For example, teachers are separately asked how the child compares "with other children in the group" during organized physical activities.

[16] A sample was selected to be representative of all Australian children in the selected age cohorts. It was drawn using a two-stage stratified sampling procedure, with the first stage selecting postcodes to ensure proportional geographic representation, and the second stage selecting children within those postcodes.

I leverage information on the development levels of other children living in the same neighbourhood (postcode) to construct a measure of the local environment. Specifically, I use the average standardized non-cognitive and cognitive development scores of similarly aged peers in the same postcode as a measure of the local reference group. For each child, I compare children of similar age (4–5 or 8–9) from both cohorts living in the same postcode. To construct this measure, I follow a two-step leave-one-out procedure. First, I de-mean interviewer-assessed development scores to account for potential cohort effects in the skill distribution by regressing the scores on a cohort indicator within each age group and retaining the residuals. Second, I compute the average de-meaned score of all other children in the same postcode, excluding the child in question. To limit sampling noise in the postcode average, I construct the measure for neighbourhoods with at least ten children across both cohorts with non-missing development scores.[17] Finally, I standardize neighbourhood scores within age groups to match the scale of individual development scores used in the analysis.

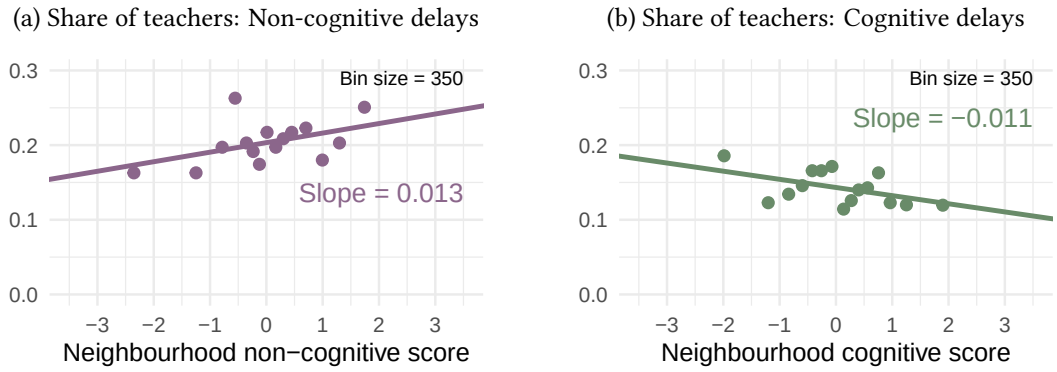
Although most of the variation in child development lies within rather than across postcodes, the share that lies across them is non-negligible, and at the observed cluster sizes the postcode averages recover most of that between-neighbourhood variation rather than sampling noise. The averages in the two domains are only weakly correlated, so a neighbourhood's standing in one domain says little about its standing in the other. This matters because studies without an independent measure of non-cognitive development have had to infer cross-neighbourhood differences in non-cognitive skill from observed differences in cognitive skill (Elder and Zhou, 2021). Measuring both domains directly eliminates the need for this assumption.[18]

As teachers' perceptions of developmental delays are informed by the child's development (see Figure 2), it can also be expected that teachers in neighbourhoods with lower average levels of child development will be more likely to report developmental delays. However, Figure 3 shows that this is not necessarily the case. Panels (a) and (b) of Figure 3 plot the average shares

[17] The results are robust to alternative methods of constructing the neighbourhood development score: using only children from the same cohort, including children of all ages, or adjusting the minimum number of observations required per postcode to ensure a stable average. See Appendix A.

[18] Appendix D reports the within- and between-postcode variance decomposition and the correlation between the two postcode averages. It also shows that postcode-average development is higher in more socioeconomically advantaged areas, and that both the between-postcode share of variance and its correlation with observed neighbourhood characteristics are somewhat larger for cognitive than for non-cognitive development.

Figure 3: Average neighbourhood development scores and teachers' perceptions at ages 4–5



Notes: Panels (a) and (b) display binned scatterplots showing the share of children perceived by teachers as having non-cognitive and cognitive delays, respectively, plotted against corresponding average neighbourhood non-cognitive and cognitive interview development scores. Both panels report raw regression lines. The sample includes children aged 4–5 attending formal care or education settings, with non-missing interview-based non-cognitive and cognitive development scores, neighbourhood average development scores, and teacher perception measures.

of children with perceived developmental delays against average neighbourhood development scores. Panel (a) reveals a positive association between neighbourhood-level non-cognitive development and teachers' likelihood of reporting non-cognitive delays. Teachers are more likely to perceive delays in neighbourhoods where children tend to score more highly on non-cognitive development. This surprising relationship may be driven by teachers having higher expectations about children's developmental milestones in neighbourhoods where most children are developmentally on track.

3 Teachers' perceptions of developmental delays

Subjective perceptions are a key source of information on children's non-cognitive skills in diagnosis, in policy, and in research.[19] Any reference group bias in these perceptions therefore distorts both measured differences in skills across groups and the identification of individual children as delayed. This section examines how teachers' perceptions vary with average neighbourhood de-

[19]Clinical guidelines direct providers diagnosing non-cognitive delays to ask parents, teachers, and other adults who care for the child about their behaviour across settings, such as at home, at school, or with peers (see CDC, "Clinical Care of ADHD"). Research relies on similar assessments: instruments such as the Strengths and Difficulties Questionnaire (SDQ) and the Behavior Problem Index (BPI) ask respondents to rate statements such as "The child often has temper tantrums or a hot temper" as "Not true", "Somewhat true", or "Certainly true". Because terms like "often" and "somewhat true" have no objective definition, respondents may supply their own standard.

velopment, conditional on the child's own measured development, and whether these perceptions differ systematically with teacher and program characteristics.

3.1 The role of the local environment in teachers' perceptions

To quantify the relationship between teachers' perceptions and the local environment, I examine how perceived developmental delays vary with the development of other children in the same neighbourhood, holding the child's own measured development fixed. The dependent variable, T_i, is an indicator equal to one if the teacher perceives that child i, aged 4–5, is less competent or much less competent than other children of a similar age in the relevant domain. Let $D_i = (D_i^n, D_i^c)'$ denote the child's interviewer-assessed non-cognitive and cognitive development scores, and let $\bar{D}_i = (\bar{D}_i^n, \bar{D}_i^c)'$ denote the corresponding leave-one-out averages among similarly aged children in the child's postcode, constructed as described in Section 2. I estimate the linear probability model

$$T_i = \beta' \bar{D}_i + \delta' D_i + \gamma' X_i + \epsilon_i \tag{1}$$

by OLS, clustering standard errors at the postcode level. The control vector X_i contains the child's gender, cohort, and age in months, together with the LSAC household socioeconomic status index, which combines parental income, education, and occupational prestige (Baker, Sipthorp, and Edwards, 2017).[20] Columns (1) and (3) of Table 1 restrict $\bar{D}_i$ and D_i to the developmental domain corresponding to the reported delay, while columns (2) and (4) include both domains.

The coefficient of interest, β, is the association between average neighbourhood development and the probability that a teacher reports a delay, conditional on the child's own measured development. This interpretation treats postcode-average development as a proxy for the local developmental environment relevant to teachers. Teachers are asked to compare each child with other children of a similar age, and the questionnaire explicitly supplies a narrower reference group whenever one is intended. If teachers apply a common age-specific standard, and if average neighbourhood development is unrelated to their reports through any other channel once own development and X_i are held fixed, then $\beta = 0$. If instead teachers evaluate children against

[20] Results are robust to using individual demographic controls, such as mother's age, marital status, education, employment, number of siblings, family income, and home language, in place of the SES index.

Table 1: Teachers' perceptions and average neighbourhood child development

	Non-cognitive delay		Cognitive delay	
	(1)	(2)	(3)	(4)
Nbrhd. non-cognitive score	0.025*	0.021*		0.014*
	(0.006)	(0.006)		(0.005)
Non-cognitive score	-0.057*	-0.043*		-0.030*
	(0.006)	(0.007)		(0.006)
Nbrhd. cognitive score		0.010	0.019*	0.016*
		(0.006)	(0.005)	(0.005)
Cognitive score		-0.052*	-0.086*	-0.082*
		(0.007)	(0.006)	(0.006)
N	5513	5251	5263	5247
Mean	0.21	0.20	0.14	0.14
R2	0.05	0.05	0.08	0.09

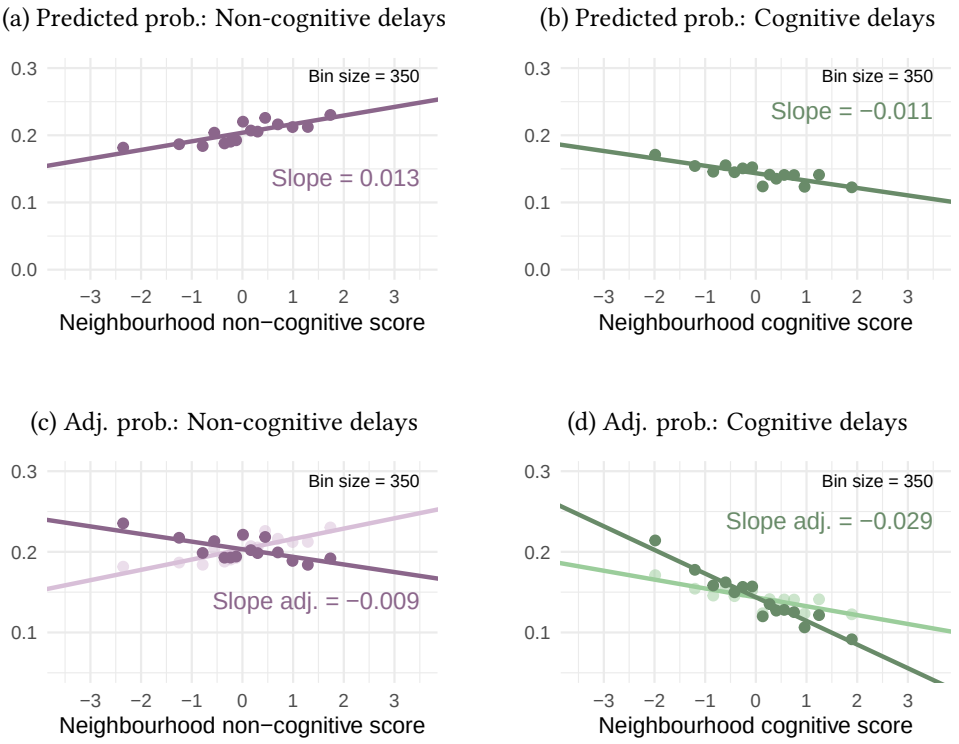
Notes: The table reports estimates from a linear probability regression using a sample of children aged 4–5 attending formal care or education settings. Controls include children's gender, cohort, and age in months, as well as a household socioeconomic status (SES) index. Standard errors are clustered at the postcode level. Significance level: * 5 percent.

those they encounter locally, a child of given measured development will be reported as delayed more often in a high-development neighbourhood, so $\beta > 0$. Because the data are observational, alternative explanations for a positive association cannot be ruled out, and the estimates cannot be interpreted as causal. Subsection 3.2 sets out these explanations and assesses each in turn.

I begin by examining how teachers' perceptions relate to children's own measured development. Table 1 shows that teachers are less likely to perceive delays in children with stronger measured development, even though they do not observe the interview-based evaluations. Comparing columns (1) and (3) with columns (2) and (4) also shows that perceptions are not entirely domain-specific. Children with stronger non-cognitive skills are less likely to be perceived as having cognitive delays, while children with stronger cognitive skills are less likely to be perceived as having non-cognitive delays.

Most importantly, Table 1 shows that the likelihood of teachers perceiving developmental delays increases with the average level of non-cognitive development in a child's neighbourhood. This pattern holds for both perceived non-cognitive and cognitive delays, even after condition-

Figure 4: Average neighbourhood development scores and predicted teachers' perceptions



Notes: Panels (a) and (b) report binned scatterplots of the predicted probabilities that children's teachers perceive developmental delays, conditional on the average neighbourhood development score (based on estimates in columns (2) and (4) of Table 1). Panel (a) shows the predicted probability of perceived non-cognitive delays conditional on the average neighbourhood non-cognitive development score, while panel (b) shows the predicted probability of perceived cognitive delays conditional on the average neighbourhood cognitive development score. Panels (c) and (d) report counterfactual predicted probabilities obtained by setting the average neighbourhood development scores to their sample means (zero), with the remaining covariates unchanged. Panel (c) displays the adjusted probability of perceived non-cognitive delays, and panel (d) displays the adjusted probability of perceived cognitive delays. All panels plot raw regression lines.

ing on children's individual development levels.[21] This implies that teachers may be less likely to recognize a delay in either domain in neighbourhoods where non-cognitive delays are more prevalent. In contrast, the average level of cognitive development in the neighbourhood predicts teacher perceptions only for cognitive delays, consistent with the findings of Kinsler and Pavan (2021) and Elder and Zhou (2021).

Next, I explore the implications of the estimated magnitudes for the role of reference groups in shaping gaps in teachers' recognition of delays between less- and more-developed areas. First,

[21] Appendix A shows that excluding the attention measure recorded during the interview from the construction of child- and neighbourhood-level non-cognitive scores weakens the relationship between average neighbourhood non-cognitive development and perceptions of cognitive delays, though the difference is not statistically significant.

I predict the probabilities of teachers perceiving delays based on estimates reported in columns (2) and (4) of Table 1. Panels (a) and (b) of Figure 4 plot these predicted probabilities against the average neighbourhood development level. The simple linear probability model replicates the patterns in Figure 3, showing a positive relationship between the predicted probability of perceiving non-cognitive delays and average neighbourhood non-cognitive development, and a weak negative relationship between the predicted probability of perceiving cognitive delays and average neighbourhood cognitive development.

I next examine how much of the observed neighbourhood pattern in perceived delays can be attributed to the local reference group under the reference-group interpretation. I construct counterfactual predicted probabilities of perceived delays, setting average neighbourhood development to the sample mean (zero) for all children. This removes the fitted component associated with neighbourhood development. Panels (c) and (d) of Figure 4 plot binned scatterplots of these adjusted probabilities against the average neighbourhood development level. Panel (c) shows that holding the reference group fixed in this way reverses the relationship between perceived non-cognitive delays and neighbourhood non-cognitive development. The relationship becomes negative, reflecting the selection of children with lower measured development and less advantaged family characteristics (as captured by control variables) into lower-development neighbourhoods.

In the data, teachers in neighbourhoods in the top quartile of average non-cognitive development are 1.1 percentage points more likely to perceive non-cognitive delays than those in bottom-quartile neighbourhoods. Once the average neighbourhood development level is held at a common value, the gap reverses. The adjusted probability of perceiving non-cognitive delays is 4.2 percentage points lower for teachers in top-quartile neighbourhoods than those in bottom-quartile neighbourhoods. A similar pattern holds for cognitive delays. Holding the reference group fixed in the same way, using the estimates in column (4) of Table 1, widens the magnitude of the negative gap in reported cognitive delays between the top and bottom quartiles of neighbourhood cognitive development from 3.0 to 9.2 percentage points.

3.2 Robustness checks

The estimates in Table 1 are descriptive and cannot by themselves establish that teachers use the local neighbourhood as a reference group. This subsection therefore examines three concerns about their interpretation. First, I assess whether the results depend on the linear functional form imposed in Equation (1). Second, I consider alternative channels through which average neighbourhood development could be related to teachers' perceptions even in the absence of local benchmarking. Third, I examine the consequences of measurement error in the interview-based development measures. Across these exercises, the positive relationship between average neighbourhood non-cognitive development and teachers' perceptions remains.

I first relax the linear functional form of the baseline specification. Column (1) of each panel in Table 2 reports average marginal effects from logit models. The estimated association between average neighbourhood non-cognitive development and perceived delays remains positive and statistically significant. Thus, the central pattern in Table 1 is not specific to the linear probability model.

A more important concern is that average neighbourhood development may capture factors other than the reference point teachers use when evaluating children. In particular, teachers and early-childhood programs may differ systematically across neighbourhoods in their qualifications, resources, or incentives to identify developmental delays. In addition, conditions during the LSAC interview may vary with neighbourhood characteristics. If children in different neighbourhoods systematically exert different effort during the assessment, or experience different levels of parental or sibling involvement, interviewer-based development scores may differ even conditional on underlying development.

To assess these possibilities, I augment Equation (1) with a detailed set of observable controls. To capture variation in interview conditions, I use measures available for the LSAC baby cohort describing parental and sibling behaviour during the cognitive assessment, including whether they were absent, remained at a distance, observed the child, encouraged the child, or interfered with the test. I additionally control for parent-reported sleeping problems. To account for differences in teachers and programs, I control for whether the teacher has a university de-

Table 2: Robustness checks

	(1) Logit AME	(2) OLS	(3) + Controls	(4) OLS	(5) TSLS
Panel A. Perceived non-cognitive delay					
Nbrhd. non-cognitive score	0.020*	0.022*	0.020*	0.016*	0.050*
	(0.006)	(0.009)	(0.009)	(0.006)	(0.013)
Non-cognitive score (baseline)	-0.037*	-0.035*	-0.034*		
	(0.005)	(0.011)	(0.011)		
Non-cognitive score (focus item)				-0.028*	-0.387*
				(0.007)	(0.111)
Nbrhd. cognitive score	0.010	0.008	0.001	0.011	0.003
	(0.006)	(0.010)	(0.012)	(0.006)	(0.018)
Cognitive score	-0.050*	-0.069*	-0.074*	-0.053*	-0.042
	(0.006)	(0.011)	(0.011)	(0.007)	(0.067)
Sample	Full	Restricted	Restricted	IV sample	IV sample
N	5251	1912	1912	5208	5208
Mean	0.20	0.19	0.19	0.20	0.20
R2		0.05	0.08	0.05	
Panel B. Perceived cognitive delay					
Nbrhd. non-cognitive score	0.013*	0.017*	0.018*	0.012*	0.025*
	(0.005)	(0.007)	(0.007)	(0.005)	(0.010)
Non-cognitive score (baseline)	-0.023*	-0.030*	-0.030*		
	(0.005)	(0.010)	(0.011)		
Non-cognitive score (focus item)				-0.029*	-0.163
				(0.006)	(0.087)
Nbrhd. cognitive score	0.017*	0.011	0.002	0.017*	0.029*
	(0.005)	(0.009)	(0.010)	(0.005)	(0.013)
Cognitive score	-0.077*	-0.084*	-0.087*	-0.083*	-0.151*
	(0.005)	(0.011)	(0.011)	(0.006)	(0.051)
Sample	Full	Restricted	Restricted	IV sample	IV sample
N	5247	1912	1912	5204	5204
Mean	0.14	0.13	0.13	0.14	0.14
R2		0.09	0.12	0.09	

Notes: The table reports estimates of Equation (1) under alternative specifications, for children aged 4–5 attending formal care or education settings. Panel A reports perceived non-cognitive delays and Panel B perceived cognitive delays. All specifications control for the child's gender, cohort, and age in months, and the household socioeconomic status (SES) index. Column (1) reports average marginal effects from a logit model on the full sample. Column (2) re-estimates the baseline linear probability model on the restricted sample for which the full set of additional controls is available, and column (3) adds these controls: conditions during the interview, teacher and program characteristics, Census neighbourhood characteristics, and Australian state/territory indicators. Columns (4) and (5) replace the baseline non-cognitive score (three-item index) with a single item, the child's degree of focus during the cognitive test, and are estimated on the sample for which all instruments are non-missing. Column (4) is estimated by OLS and column (5) by two-stage least squares, instrumenting the focus item with the positive- and negative-behaviour ratings recorded during the same interview and the PPVT score with the WAI score. Standard errors are clustered at the postcode level. Significance level: * 5 percent.

gree (versus a diploma or certificate), an indicator measuring teachers' experience in childcare (0–5 years versus 6 or more years), whether the child is attending daycare (versus preschool or kindergarten), the age range of the children's class reported by the teacher, and the ratio of children to qualified staff. Subsection 3.3 examines these characteristics in greater detail.[22] Finally, I add Census-based neighbourhood characteristics that may proxy for differences in the composition of local populations or in resources available for identifying and supporting children with delays.[23] These include the shares of children aged 0–4 and 5–9, the shares of individuals of Aboriginal origin, who speak English at home, and who were born in Australia, as well as the Index of Relative Socioeconomic Advantage and Disadvantage. The latter summarizes a broad range of neighbourhood characteristics related to access to material and social resources and opportunities to participate in society (Australian Bureau of Statistics, 2011).[24] I also include Australian state/territory indicators to account for regulatory differences across early-childhood education systems.

Column (2) of Table 2 re-estimates Equation (1) on the restricted sample for which the full set of additional controls is available, while column (3) adds these controls. The estimated relationships between perceived delays and average neighbourhood non-cognitive development change little either when restricting the sample or when adding the controls. These observable differences therefore do not account for the baseline pattern.

I next examine measurement error in the interview-based development scores. Because children in the same postcode resemble one another, the neighbourhood average $\bar{D}_i$ may carry information about a child's own underlying development that the noisy individual measure D_i fails to capture. Importantly, this mechanism cannot generate the positive neighbourhood coefficient. Children with higher underlying development are less likely to be perceived as delayed,

[22] See Appendices B.5 and B.7 for summary statistics and additional discussion.

[23] See Appendix B.3 for summary statistics on neighbourhood characteristics.

[24] One alternative interpretation of the positive neighbourhood coefficient is that teachers in better-resourced neighbourhoods have stronger incentives to identify developmental delays, for example because more support services are available once a delay is recognized. The added controls account for observable differences in neighbourhood resources that may generate such incentives. As an additional check, Appendix E re-estimates Equation (1) using an indicator for perceived advanced development instead of developmental delays. Differences in incentives to identify children requiring support should primarily affect reports of delays, whereas reference-dependent evaluation should also affect judgments of advanced development. Conditional on the child's own development, children in neighbourhoods with higher average non-cognitive development are significantly less likely to be perceived as advanced (Table E.2), as predicted by reference-dependent evaluation.

so any component of the child's own development captured by $\bar{D}_i$ induces a downward bias in its coefficient. Classical measurement error in $\bar{D}_i$ itself similarly attenuates the neighbourhood coefficient toward zero. Both mechanisms therefore work against, rather than generate, the positive relationship observed in the data.[25]

To assess attenuation from measurement error in the individual development measures, I estimate a two-stage least squares version of Equation (1) (Kinsler and Pavan, 2021; Agostinelli and Wiswall, 2025). Column (4) of Table 2 re-estimates the equation by OLS on the sample for which all instruments are available, replacing the baseline non-cognitive development score with the child's degree of focus during the cognitive assessment. Column (5) estimates the same specification on the same sample by two-stage least squares, instrumenting the focus measure with the positive- and negative-behaviour ratings recorded during the same interview and the PPVT score with the WAI score. All measures are standardized by age.[26]

The adjusted estimates increase the magnitude of the relationship between teachers' perceptions and the child's own measured development and, more importantly, increase rather than eliminate the coefficient on average neighbourhood non-cognitive development. Thus, under the maintained classical error assumption used in this exercise, accounting for attenuation strengthens the positive neighbourhood relationship.[27]Appendix F extends the analysis by also allowing for measurement error in the neighbourhood development measures. Both the TSLS and GMM estimates again yield larger, though less precise, neighbourhood coefficients.

A distinct concern is that measurement error may be systematic rather than classical. In particular, interviewer assessments of non-cognitive skills could vary with neighbourhood characteristics conditional on the child's household, causing $\bar{D}_i$ to partly reflect differences in assessment rather than differences in child development. Interviewer identifiers are not available in LSAC, so this possibility cannot be addressed directly using interviewer fixed effects. The

[25]Kinsler and Pavan (2021) make an analogous argument for noise in measures of children's cognitive skill. Their dependent variable indicates that a parent believes the child is above average, so the corresponding prediction in their setting has the opposite sign.

[26]Because these specifications replace the baseline non-cognitive development score based on three items with a single item, the coefficient on individual non-cognitive development is not directly comparable in levels with the estimates in columns (1)–(3).

[27]The substantial attenuation is consistent with the measurement-error estimates for non-cognitive skill measures in the previous literature. For example, Cunha et al. (2010) estimate signal shares of 1–61% for multi-item measures of children's non-cognitive skills, while individual non-cognitive survey items can contain substantially less signal.

additional-controls specifications nevertheless provide evidence against the observable version of this explanation. On a common sample, adding the Index of Relative Socioeconomic Advantage and Disadvantage, Census composition measures, and controls for conditions during the interview has little effect on the estimated neighbourhood relationship. Thus, systematic differences in assessments associated with these observed neighbourhood characteristics do not account for the estimated relationship.

Taken together, these exercises strengthen the interpretation of the positive neighbourhood relationship as reflecting variation in teachers' local reference points. The relationship remains stable across the alternative explanations and measurement concerns examined here. The next subsection evaluates whether teachers' ability to identify children with low measured development varies systematically with their qualifications and classroom environments.

3.3 Teacher and program characteristics and delay identification

There is strong evidence that high-quality early childhood education improves the developmental outcomes of disadvantaged children.[28] Complementing this evidence, a large body of research examines how teacher qualifications and classroom characteristics influence the quality of the classroom environment and, in turn, affect student outcomes.[29] One potential mechanism is the teachers' ability to identify developmental delays and provide appropriate support. In this subsection, I examine whether teacher qualifications, measured by education or experience, and classroom characteristics, including childcare setting type, child-staff ratio, or age composition, are associated with the identification of children with low measured levels of development.

During the 2000s, early childhood education and care in Australia was governed by diverse jurisdictional arrangements with no nationally agreed staffing standards (Elliott, 2006; Warren and Haisken-DeNew, 2013).[30] Early childhood staff could qualify through either a university degree or a diploma or certificate from a Registered Training Organisation. Around 37 percent of

[28] See Currie and Almond (2011) and Elango et al. (2015) for surveys.

[29] For example, Chetty et al. (2011) find that having a more experienced teacher in kindergarten is associated with higher earnings in adulthood. Manning et al. (2019) show that teachers' educational attainment is a consistent and important predictor of early childhood education quality.

[30] Australia's major reform aimed at standardizing the quality of early childhood education and care, the National Quality Framework, came into effect in 2012, after the period when children in my baseline sample were aged 4–5. Therefore, it is unlikely to affect the results presented in this section.

educators in my sample did not hold a university degree.[31] Children aged 4–5 attended a range of settings, mostly preschool or kindergarten, with 26 percent in daycare (see Appendix B.6). This variation in qualifications and settings lets me examine whether training is associated with the recognition of developmental delays.

For each dimension of development, I split the sample into subsamples with low and high measured development based on interview-based development scores. The subsample of children with low measured non-cognitive (or cognitive) development consists of those whose interview-based development score falls within the first quartile of the distribution for children aged 4–5. Similarly, the subsample of children with high measured development includes those with scores in the top quartile. Intuitively, children with the lowest measured development are the group for whom developmental concerns are more plausible and for whom recognition is especially relevant.[32]

Next, I estimate the linear probability regression separately for the two subsamples, $J = \{H,L\}$:

$$T_i = \beta^{J,V\prime} V_i + \gamma^{J,X\prime} X_i + \epsilon_i, \tag{2}$$

where X_i is the control vector from Equation (1) and V_i collects observed teacher and classroom characteristics that may be associated with the quality of early childhood education.[33] Two of these describe the teacher: whether she holds a university degree (versus a diploma or certificate) and whether she has 0–5 years of experience in childcare (versus 6 or more). The remainder describe the setting: whether the child attends daycare (versus preschool or kindergarten), the ages of the youngest and oldest children in the class as reported by the teacher, and the child-to-qualified-staff ratio.[34]

[31] See Appendix B.7 for the summary of teacher qualifications.

[32] For example, in 2023, 24.2 percent of students in Australian schools were offered accommodations due to predominantly non-cognitive and cognitive deficits. The results in this subsection are robust to dividing children into subsamples with measured development above and below the sample median score.

[33] These are the same teacher and classroom characteristics used as controls in column (3) of Table 2. There, they control for variation in program quality, whereas here their coefficients are the object of interest and neighbourhood average development is excluded.

[34] In the 2000s, naming conventions and starting ages for early childhood programs varied across Australian states and territories. Preschool typically referred to the year before formal schooling, while kindergarten marked the first year of full-time school, though terminology differed by jurisdiction (Australian Bureau of Statistics, 2004). Due to inconsistencies in reporting by parents regarding preschool versus kindergarten attendance (Australian Institute of Family Studies, 2025), I combine these categories in the baseline analysis. The association between teacher education

Table 3: Relationship between teacher and program characteristics and perceived delays

	Non-cognitive delay		Cognitive delay	
	(1)	(2)	(3)	(4)
	Non-cogn. score low	Non-cogn. score high	Cogn. score low	Cogn. score high
Univ. degree	0.085*	0.020	0.056*	-0.001
	(0.027)	(0.021)	(0.027)	(0.014)
Experience 0-5 years	-0.091*	0.005	0.018	0.041
	(0.032)	(0.028)	(0.035)	(0.021)
Age of youngest	0.003	0.001	0.003	0.002*
	(0.002)	(0.001)	(0.002)	(0.001)
Age of oldest	0.001	0.001	0.001	0.001
	(0.002)	(0.001)	(0.002)	(0.001)
Child to staff ratio	-0.002	0.002	0.002	0.001
	(0.002)	(0.002)	(0.002)	(0.001)
Child attends daycare	-0.050	-0.014	-0.025	0.001
	(0.033)	(0.026)	(0.029)	(0.015)
N	1364	1407	1213	1484
Mean	0.30	0.16	0.29	0.06
R2	0.07	0.03	0.05	0.03

Notes: The table reports estimates from a linear probability regression for low-development (measured development bottom quartile) and high-development (measured development top quartile) samples, limited to children aged 4–5 attending formal care or education settings. Controls include children's gender, cohort, and age in months, as well as the household socioeconomic status (SES) index. Standard errors are clustered at the postcode level. Significance level: * 5 percent.

The results of this specification are presented in Table 3. Among children with low measured levels of development, teacher education is positively associated with delay recognition. Specifically, teachers with bachelor's or postgraduate university degrees are 8.5 percentage points more likely to report non-cognitive delays and 5.6 percentage points more likely to report cognitive delays than those with diplomas or certificates. This relationship is not evident among children with high measured development, for whom more extensive training should not increase the likelihood of perceiving delays. Among children with low measured non-cognitive development, less-experienced teachers (those with 0–5 years of experience) are 9.1 percentage points less likely to perceive non-cognitive delays compared to those with 6 or more years of experience. This experience-related gap is not apparent for cognitive delays. Other classroom characteristics, such as the child-to-staff ratio and age composition, are not robustly associated with delay recognition.

and delay recognition remains robust when distinguishing between children attending preschool and kindergarten (see Table B.11).

Having shown that teacher education and experience are associated with delay recognition among children with low measured development, I next examine whether these qualifications are also associated with two distinct features of teachers' perceptions. The first is the own-development alignment margin, which captures how closely perceptions correspond to the child's independently measured development. The second is the reference-group margin, which captures how perceptions vary with average neighbourhood development. Table 4 reports estimates from Equation (1), interacting the child and neighbourhood development scores with indicators for teacher qualifications. Columns (1) and (4) interact these measures with an indicator for whether the teacher holds an undergraduate or postgraduate degree, columns (2) and (5) with an indicator for 0–5 years of experience, and columns (3) and (6) with both. All regressions control for the full set of teacher and classroom characteristics.

On the own-development alignment margin, both qualifications are associated with how closely perceptions track children's measured development, though experience matters only in the non-cognitive domain. The interaction between the child's cognitive score and the teacher-degree indicator is negative and statistically significant in columns (1) and (4), and changes little when the experience interactions are added in columns (3) and (6). University-trained teachers' perceptions of both non-cognitive and cognitive delays are roughly one and a half to two times as strongly associated with measured cognitive development as those of teachers with diplomas or certificates. For experience, the interaction with the child's non-cognitive score is positive and statistically significant in columns (2) and (3). Because the baseline relationship is negative, this implies weaker alignment among less-experienced teachers. No corresponding relationship with experience appears for perceived cognitive delays.

On the reference-group margin the two qualifications diverge. The interactions between the teacher-degree indicator and average neighbourhood development are small and statistically insignificant in both domains, so there is little evidence that university training is associated with weaker local benchmarking. Experience, by contrast, is associated with the reference-group margin. In columns (2) and (3), the interaction between 0–5 years of experience and average neighbourhood non-cognitive development is positive and statistically significant, indicating a stronger neighbourhood gradient among less-experienced teachers. As on the alignment margin,

Table 4: Teachers' perceptions and qualifications

	Non-cognitive delay			Cognitive delay		
	(1) By degree	(2) By exp.	(3) Both	(4) By degree	(5) By exp.	(6) Both
Non-cognitive score	-0.038*	-0.053*	-0.042*	-0.035*	-0.036*	-0.036*
	(0.011)	(0.007)	(0.011)	(0.011)	(0.007)	(0.011)
× univ. degree	-0.015		-0.018	0.001		0.001
	(0.015)		(0.015)	(0.014)		(0.014)
× exp. 0–5		0.036*	0.036*		0.009	0.007
		(0.017)	(0.017)		(0.017)	(0.017)
Cognitive score	-0.036*	-0.054*	-0.037*	-0.063*	-0.088*	-0.067*
	(0.011)	(0.008)	(0.012)	(0.009)	(0.007)	(0.009)
× univ. degree	-0.028*		-0.027	-0.034*		-0.035*
	(0.014)		(0.014)	(0.012)		(0.012)
× exp. 0–5		-0.005	-0.003		0.020	0.024
		(0.015)	(0.015)		(0.016)	(0.016)
Nbrhd. non-cognitive score	0.016	0.018*	0.012	0.023*	0.016*	0.021*
	(0.010)	(0.007)	(0.010)	(0.008)	(0.006)	(0.009)
× univ. degree	0.011		0.008	-0.008		-0.009
	(0.013)		(0.013)	(0.011)		(0.011)
× exp. 0–5		0.028*	0.029*		0.010	0.012
		(0.013)	(0.013)		(0.012)	(0.012)
Nbrhd. cognitive score	0.003	0.002	0.002	0.009	0.013*	0.008
	(0.009)	(0.007)	(0.010)	(0.008)	(0.006)	(0.008)
× univ. degree	0.004		0.000	0.009		0.007
	(0.014)		(0.014)	(0.011)		(0.011)
× exp. 0–5		0.024	0.025		0.011	0.011
		(0.015)	(0.015)		(0.016)	(0.016)
N	4564	4564	4564	4561	4561	4561
R2	0.07	0.07	0.07	0.11	0.11	0.11

Notes: The table reports estimates from linear probability regressions of perceived delays on child and neighbourhood development measures, interacted with indicators for teacher qualifications, for children aged 4–5 attending formal care or education settings. Columns (1) and (4) interact these measures with an indicator for whether the teacher holds a university degree; columns (2) and (5) with an indicator for 0–5 years of experience in childcare; and columns (3) and (6) with both. Controls include the child's gender, cohort, and age in months; a household socioeconomic status (SES) index; an indicator for whether the teacher has a university degree (versus a diploma or certificate); whether the child attends daycare (versus preschool or kindergarten); the age range of the children's class reported by the teacher; the ratio of children to qualified staff; and an indicator for teachers' experience in childcare (0–5 years versus 6 or more years). Standard errors are clustered at the postcode level. Significance level: * 5 percent.

the pattern is specific to non-cognitive delays.[35]

These findings suggest that teacher qualifications are associated not only with whether delays are perceived, but also with how teachers' perceptions relate to children's measured development and the local environment. These differences in teachers' perceptions may have consequences beyond teachers' own assessments because professional judgment plays a central role in decisions about how children are supported. For example, the *Early Years Learning Framework*, published by the Australian Government Department of Education, highlights the role of teachers' assessments in planning learning activities, communicating progress, monitoring advancement toward learning outcomes, and identifying children who need additional support (Australian Government Department of Education, Employment and Workplace Relations, 2009). Teachers' evaluations may also shape the information available to families and subsequent use of developmental support services. In the following sections, I examine whether teachers' perceptions are reflected in mothers' beliefs and whether teachers' and mothers' perceptions predict subsequent therapy uptake.

4 The relationship between teachers' and mothers' perceptions

Teachers are one of the main sources parents rely on to monitor their children's progress. This section asks whether mothers' perceptions vary with average neighbourhood development in the way teachers' do, which would indicate that mothers also benchmark against local peers, and whether they are associated with the assessments teachers and schools convey to them.

I estimate variants of the following linear probability model separately at ages 4–5 and 8–9, restricting the sample to the roughly 95 percent of households in which the mother is the primary respondent. The dependent variable $M_{i,t}$ equals one if the mother reports that her child at age t

[35] As a complementary check, Appendix G estimates Equation (1) separately for subsamples of children defined by their teacher's education and experience, allowing all coefficients to differ across groups. The resulting coefficient-equality tests yield the same qualitative conclusions as Table 4.

is more difficult than other children of a similar age:

$$M_{i,t} = \beta^{MN\prime}\bar{D}_{i,t} + \beta^{MT\prime}T_{i,t} + \delta^{M\prime}D_{i,t} + \gamma^{MX\prime}X^{M}_{i,t} + \phi^{M}M_{i,t-1} + \epsilon^{M}_{i,t}. \tag{3}$$

The vectors $\bar{D}_{i,t}$ and $D_{i,t}$ contain the average neighbourhood and the child's own development scores, and $T_{i,t}$ contains the assessments available to the mother at age t, whose content differs across the two ages as described below. Specifications that include $T_{i,t}$ also condition on the mother's prior perception $M_{i,t-1}$, measured before the teacher's report, to account for persistent unobserved differences in parental perceptions, such as skill valuation, optimism, or involvement.[36] The controls $X^{M}_{i,t}$ are those of Equation (1) plus the mother's depression score and, at ages 8–9, her school involvement.[37] At each age, I estimate three specifications. The first includes the neighbourhood averages and the child's own development, the second adds teachers' perceptions and the mother's prior belief, and the third adds neighbourhood controls.

The content of $T_{i,t}$ differs by age, and this difference changes the interpretation of β^{MT}. At ages 4–5, $T_{i,t}$ captures the teacher's private perceptions of non-cognitive and cognitive delays analyzed in Section 3, which the mother need not observe, so β^{MT} reflects the conditional association between the teacher's and the mother's reports. Teachers rate children only at ages 4–5, so at ages 8–9 I instead use an indicator for whether the school contacted the mother about her child's behaviour in the previous year, which is a signal the mother directly observes. Here β^{MT} measures the association between that contact and the mother's perception, conditional on the belief she held two years earlier.

Columns (1) and (4) of Table 5 report the counterpart of the teacher specification in Table 1. Mothers' perceptions are unrelated to average neighbourhood non-cognitive development at either age. Mothers therefore do not seem to benchmark against the local non-cognitive reference group in the way teachers do. A smaller association working through teachers' reports is not ruled out, but neither would it be detectable here.[38] Mothers' perceptions are positively

[36] Since the kindergarten cohort is first observed at ages 4–5, this specification can only be estimated for the baby cohort, followed from ages 0–1.

[37] I control for maternal school involvement because variation in school contact may partly reflect schools' propensity to communicate with more or less engaged mothers, and because more engaged mothers may also have greater exposure to the child's peers, either of which can move maternal perception independently of the child's development.

[38] Combined with the teacher-to-mother association of 0.084 in column (2), the teacher gradient of 0.021 in Table 1

Table 5: The relationship between mothers' perceptions of non-cognitive delay, average neighbourhood child development, and teacher perceptions

	Ages 4–5			Ages 8–9		
	(1)	(2)	(3)	(4)	(5)	(6)
Nbrhd. non-cognitive score	-0.001	-0.003	-0.005	0.003	0.003	0.002
	(0.006)	(0.005)	(0.006)	(0.004)	(0.003)	(0.004)
Nbrhd. cognitive score	0.013*	0.013*	0.004	-0.003	-0.006	-0.005
	(0.006)	(0.006)	(0.008)	(0.005)	(0.004)	(0.004)
Teacher: Non-cognitive delay		0.084*	0.084*			
		(0.023)	(0.023)			
Teacher: Cognitive delay		0.015	0.015			
		(0.026)	(0.026)			
School contact about behaviour					0.107*	0.108*
					(0.016)	(0.016)
Non-cognitive score	-0.028*	-0.023*	-0.023*	-0.027*	-0.016*	-0.016*
	(0.009)	(0.008)	(0.008)	(0.006)	(0.005)	(0.005)
Cognitive score	-0.007	0.002	-0.000	-0.009	-0.003	-0.003
	(0.008)	(0.008)	(0.008)	(0.005)	(0.004)	(0.004)
Lag Moth.: Non-cognitive delay		0.341*	0.341*		0.517*	0.517*
		(0.049)	(0.048)		(0.031)	(0.031)
Nbrhd. controls	No	No	Yes	No	No	Yes
N	1634	1634	1634	3924	3924	3924
Mean	0.06	0.06	0.06	0.07	0.07	0.07
R2	0.02	0.15	0.15	0.03	0.29	0.29

Notes: The table reports estimates from linear probability regressions using a sample of children observed at ages 4–5 and 8–9 who attend formal care or education settings and whose mother responds to the face-to-face interview. Lagged variables are measured at ages 2–3 and 6–7, respectively. Controls in all specifications include the child's gender, cohort (for children aged 8–9), age in months, the household socioeconomic status (SES) index, the mother's depression score, and the mother's school involvement (for children aged 8–9). Standard errors are clustered at the postcode level. Significance level: * 5 percent.

associated with average neighbourhood cognitive development at ages 4–5, but not at ages 8–9. At ages 4–5, however, this association appears to reflect broader neighbourhood advantage. Adding neighbourhood controls in column (3) reduces the coefficient and renders it statistically insignificant.

Columns (2) and (5) add teachers' perceptions and the mother's prior belief. Mothers of children whose teachers report a non-cognitive delay are 8.4 percentage points more likely to report one themselves, against a base rate of 6 percent, even conditional on the belief they held two years earlier. This relationship does not extend across domains, as teachers' perceptions of cognitive delay are not associated with mothers' non-cognitive assessments. These estimates do not establish the direction of information flow, which could also run from mothers to teachers.[39] At ages 8–9 the signal is one the mother directly observes. Mothers contacted by the school about their child's behaviour in the past year are 10.7 percentage points more likely to perceive their child as more difficult, against a base rate of 7 percent. Adding neighbourhood controls in columns (3) and (6) has little effect on these associations.

While teachers' evaluations of academic progress are known to inform parents (Dizon-Ross, 2019; Doss et al., 2019), their role in shaping parental beliefs about non-cognitive development is less understood. Overall, the results imply that average neighbourhood child development has little direct relationship with mothers' perceptions, whereas teacher-reported delays and school-initiated contact about behaviour both strongly predict mothers' beliefs even conditional on prior maternal perceptions. This pattern is consistent with teachers' and schools' assessments of non-cognitive development being reflected in how mothers evaluate their own children, so that the accuracy of these assessments can matter for families and for the investments they make in their children. Section 5 examines whether these perceptions are associated with the professional services children subsequently receive.

implies an indirect relationship of approximately 0.002, about a third of the standard error in column (1). Appendix H shows that the absence of an association is also not accounted for by classical measurement error in the child's development scores.

[39] For example, mothers of children diagnosed with Autism Spectrum Disorder or ADHD may proactively inform teachers of their child's needs.

5 The relationship between perceptions and therapy uptake

Therapy uptake is measured from mothers' reports of professional services the child used in the previous 12 months: non-cognitive therapy equals one if the child saw a guidance counsellor or used another psychological or behavioural service, and cognitive therapy equals one if the child received speech therapy.[40] I estimate a linear probability model in which the dependent variable equals one if the child receives therapy ($I^S_{i,t} \in \{0,1\}$) at ages 6–7:

$$I^S_{i,t} = \beta^{SN\prime}\bar{D}_{i,t-1} + \beta^{ST\prime}T_{i,t-1} + \beta^{SM\prime}M_{i,t-1} + \delta^{S\prime}D_{i,t-1} + \phi^S I^S_{i,t-1} + \gamma^{SX\prime}X^S_{i,t} + \epsilon^S_{i,t}, \quad (4)$$

where $\bar{D}_{i,t-1}$ and $D_{i,t-1}$ are the average neighbourhood and own non-cognitive and cognitive development scores at ages 4–5. The controls $X^S_{i,t}$ are those of Equation (1), measured at ages 6–7, plus Census neighbourhood characteristics and Australian state/territory indicators, which proxy for variation in the supply of therapy services. Perceptions and development enter with a one-wave lag, so they are measured before the outcome and cannot respond to therapy taken up at ages 6–7. I additionally control for prior therapy uptake at ages 4–5 ($I^S_{i,t-1}$), following a value-added approach that aims to account for individual heterogeneity and the history of past inputs (Todd and Wolpin, 2007; Fiorini and Keane, 2014). All specifications are estimated on a common sample.

Columns (1) and (4) of Table 6 relate uptake to the neighbourhood averages and the child's own development alone. A one standard deviation increase in average neighbourhood non-cognitive development at ages 4–5 is associated with a 1.6 percentage point higher probability of receiving cognitive therapy, against a base rate of 8 percent, and a 0.9 percentage point higher probability of non-cognitive therapy, against a base rate of 5 percent. Neither outcome is associated with average neighbourhood cognitive development.

Columns (2) and (5) add teachers' and mothers' perceptions and prior therapy use. Both predict uptake in the corresponding developmental domain. Children whose mothers perceive a non-cognitive delay are 17.6 percentage points more likely to receive non-cognitive therapy, and those whose teachers do are 2.5 points more likely. For cognitive therapy, teachers' perceptions

[40] Summary statistics for the full sample appear in Appendix B.4.

Table 6: The relationship between therapy uptake, average neighbourhood child development, and perceptions

	Non-cognitive			Cognitive		
	(1)	(2)	(3)	(4)	(5)	(6)
Teach.: Non-cognitive delay		0.025*	0.022		0.006	0.011
		(0.012)	(0.014)		(0.014)	(0.016)
Teach.: Cognitive delay		0.003	0.007		0.065*	0.064*
		(0.016)	(0.018)		(0.020)	(0.023)
Moth.: Non-cognitive delay		0.176*	0.170*		0.038	0.042
		(0.030)	(0.035)		(0.022)	(0.026)
Moth.: Concern cognitive delay		0.032	0.027		0.063*	0.063*
		(0.022)	(0.026)		(0.026)	(0.031)
Nbrhd. non-cognitive score	0.009*	0.006		0.016*	0.016*	
	(0.004)	(0.004)		(0.004)	(0.004)	
Nbrhd. cognitive score	0.003	0.002		0.008	0.006	
	(0.005)	(0.005)		(0.006)	(0.006)	
Non-cognitive score	-0.007	0.001	-0.002	-0.020*	-0.012*	-0.010
	(0.005)	(0.004)	(0.005)	(0.006)	(0.005)	(0.007)
Cognitive score	-0.003	0.002	0.003	-0.040*	-0.024*	-0.025*
	(0.004)	(0.004)	(0.005)	(0.006)	(0.006)	(0.007)
Therapy at 4–5		0.117*	0.111*		0.222*	0.222*
		(0.039)	(0.045)		(0.021)	(0.025)
Nbrhd. controls	Yes	Yes	No	Yes	Yes	No
Nbrhd. FE	No	No	Yes	No	No	Yes
N	3606	3606	3606	3606	3606	3606
Mean	0.05	0.05	0.05	0.08	0.08	0.08
R2	0.01	0.07	0.26	0.05	0.15	0.29

Notes: The table reports estimates from linear probability regressions, for children aged 6–7 attending formal care or education settings, with the mother as the face-to-face respondent. Perceptions and development are measured at ages 4–5. All columns control for the child's gender, cohort and age in months; the household SES index; and the child's own non-cognitive and cognitive scores at ages 4–5. Columns (1), (2), (4), and (5) additionally include Census neighbourhood characteristics and Australian state/territory dummies, which columns (3) and (6) absorb through postcode fixed effects. Columns (2), (3), (5), and (6) also control for lagged therapy uptake. Standard errors are clustered at the postcode level. Significance level: * 5 percent.

of a cognitive delay are associated with a 6.5 percentage point increase and mothers' cognitive concerns with a comparable 6.3 points. In each case perceptions in the other domain are unrelated to uptake. Prior therapy use is also strongly associated with subsequent uptake, consistent with persistence in the use of these services. The coefficient on average neighbourhood non-cognitive development is essentially unchanged for cognitive therapy, and falls and loses statistical significance for non-cognitive therapy. The perceptions measured here therefore account for little of the relationship between the local environment and uptake.[41]

Columns (3) and (6) replace the neighbourhood-level averages and controls with postcode fixed effects, so that the perception coefficients are identified only from comparisons among children living in the same postcode. The estimates remain similar: 6.4 and 6.3 percentage points for cognitive therapy, and 17.0 percentage points for mothers' perceptions of non-cognitive delay. The exception is teachers' perceptions of non-cognitive delay, for which the estimate falls to 2.2 percentage points and is no longer statistically significant.

Perceptions therefore predict which children receive professional support, conditional on measured development and prior use, and within postcodes.[42] Whether a delay is recognized is consequential for the individual child, not only for how areas compare in reported delay rates. Perceived delays are also associated with other forms of family investment. Appendix J examines whether mothers' perceptions of non-cognitive delay at ages 8–9 predict measures of the home environment at ages 10–11, conditional on the corresponding home-environment measure at ages 8–9 and other child, family, and neighbourhood characteristics. Mothers who perceive a delay are more likely to report using parenting education programs, support groups, and tutoring.[43]

[41] The estimated magnitudes imply as much. The teacher gradient of 0.021 in Table 1, combined with the teacher-to-therapy association of 0.025, delivers an indirect relationship of roughly 0.0005, and mothers' perceptions of non-cognitive delay show no neighbourhood gradient at all (see Table 5).

[42] Appendix I shows little evidence of heterogeneity in these associations across more- and less-educated households.

[43] They also report lower educational aspirations for their children, lower warmth, and higher anger, though these associations cannot separate a response to the perception from the child behaviour that produces it.

6 Conclusion

Using direct interviewer observations of non-cognitive development, I show that, conditional on children's standardized development measures, teachers are less likely to report perceived delays in neighbourhoods with lower average measured non-cognitive development. As a result, neighbourhood differences in teacher-reported delays substantially understate the differences implied by the standardized development measures.

These perception differences matter beyond their implications for measurement. Teachers' and mothers' perceptions of developmental delays predict subsequent therapy uptake in the corresponding developmental domain, even conditional on children's measured development, prior therapy use, and unobserved postcode characteristics. Perceived delays are therefore associated with the professional support children subsequently receive, while the perceptions themselves depend in part on where children live.

Taken together, the findings suggest the value of more consistent approaches to identifying and evaluating non-cognitive delays in early childhood. Policy initiatives such as the CDC's *Learn the Signs. Act Early.* program can help improve awareness of age-appropriate developmental milestones among both early childhood education professionals and families. Supplementing subjective perceptions with standardized assessment tools, such as direct observations guided by structured rating scales, would offer a valuable addition to current assessment practices.

In parallel, university-educated early childhood teachers are more likely to report perceived delays among children with low measured development, and more experienced teachers are more likely to do so for non-cognitive delays. This makes teacher training and qualifications a potentially important margin for policy and supports policies aimed at improving the consistency of educational requirements for early childhood education professionals, such as the National Quality Framework in Australia.

Data availability

The data used in this study are from *Growing Up in Australia: the Longitudinal Study of Australian Children (LSAC)*. The data are available to researchers through the Australian Data Archive subject to the LSAC conditions of access. The dataset used in this study is Department of Social Services, Australian Institute of Family Studies, and Australian Bureau of Statistics (2022), "Growing Up in Australia: Longitudinal Study of Australian Children (LSAC) Release 9.1 C2 (Waves 1–9C)," DOI: 10.26193/QR4L6Q.

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CRediT authorship contribution statement

Anastasiia Suvorova: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to edit prose for readability and to help organize the structure of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Declaration of competing interest

The author declares that she has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Online Appendix to

"(Mis)perceptions about children"

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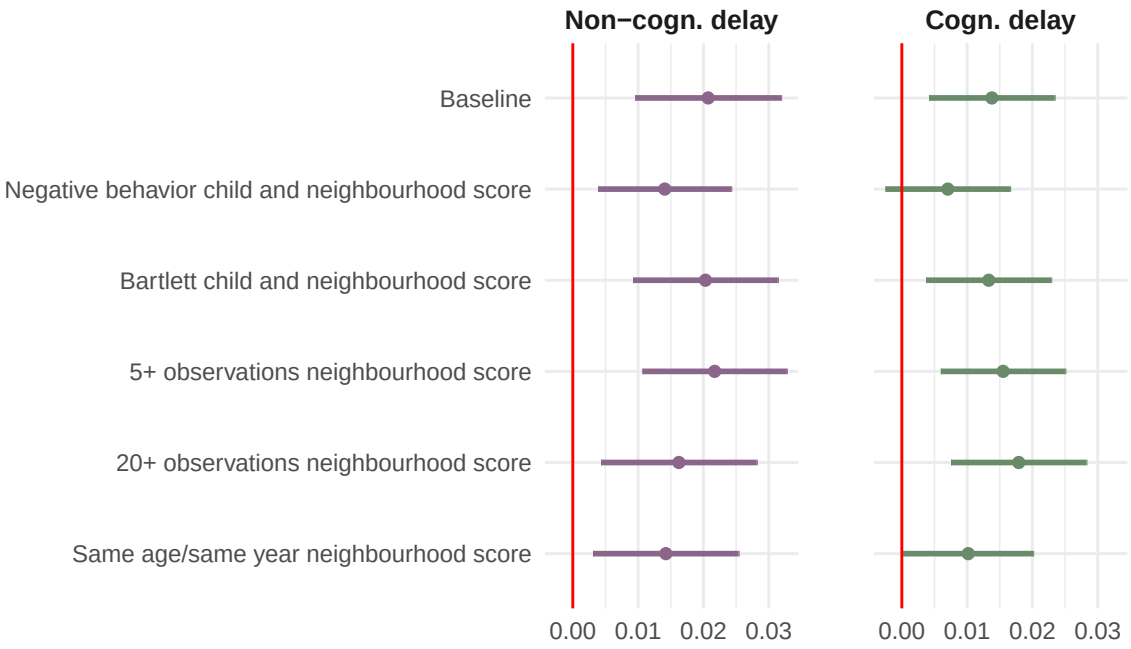
A Robustness to the construction of average neighbourhood development

This section examines the sensitivity of the estimates reported in columns (2) and (4) of Table 1 to alternative methods for constructing individual non-cognitive skill scores and average neighbourhood child development scores. Figure A.1 presents estimates of the coefficient on average neighbourhood non-cognitive development from Equation (1), using various measurement approaches.

- Baseline: Replicates the estimates reported in Columns (2) and (4) of Table 1.
- Negative child and neighbourhood score: Uses only the degree of negative behaviour during the interview as the measure of non-cognitive skills, excluding indicators of focus during cognitive testing and positive behaviours. Neighbourhood averages are computed following the method in Subsection 2.3.
- Bartlett child and neighbourhood score: Constructs non-cognitive skill scores using Bartlett factor scores from three interviewer-assessed items. Neighbourhood averages follow the same approach as above.
- 5+ observations neighbourhood score: Retains the baseline individual score but relaxes the requirement for computing neighbourhood averages to at least five observations of non-cognitive skills from children other than the index child, across both cohorts.
- 20+ observations neighbourhood score: Uses the baseline individual score but requires at least 20 observations to compute neighbourhood averages, imposing a stricter data threshold than the baseline (10+).
- Same age/same year neighbourhood score: Uses the baseline individual score but restricts the neighbourhood average to children from the same cohort (i.e., similar age and the same survey wave), requiring at least five observations per postcode. The baseline, by contrast, pools data from both cohorts (similar age, different waves) after demeaning.

Across specifications, the estimated association between average neighbourhood non-cognitive development and teacher perceptions remains stable. Relying solely on negative behavioural re-

Figure A.1: Estimated coefficient on the average neighbourhood non-cognitive development - alternative non-cognitive score construction methods



Notes: This figure shows the estimated coefficient on average neighbourhood non-cognitive development along with 95 percent confidence intervals from linear regressions of perceived delays on child and neighbourhood cognitive and non-cognitive development levels based on Equation 1 using alternative methods of constructing non-cognitive scores. All models control for child gender, cohort, age in months, and household socioeconomic status (SES) index. Standard errors are clustered at the postcode level. The sample is limited to children aged 4–5 who attended formal care or education settings.

sponses reduces the estimated association with perceived cognitive delays, but the confidence intervals overlap with the baseline.

B Additional data description

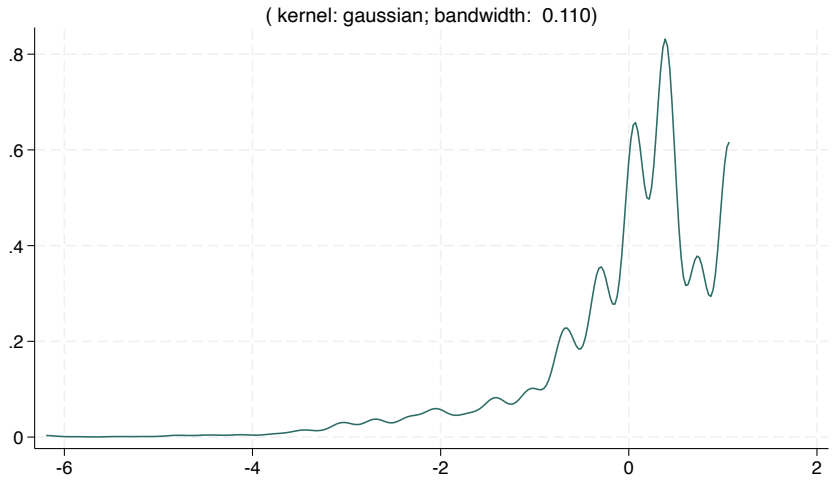
B.1 Measures of child development and perceptions

Table B.1: Principal component analysis loadings by age group

	Aged 4–5	Aged 8–9
Positive behaviour	0.47	0.60
Negative behaviour	0.63	0.67
Focus during PPVT	0.62	0.44
Eigenvalue (PC1)	1.24	1.12
Number of Obs.	9180	8239

Notes: The table displays the loadings from a principal component analysis (PCA) of direct observation measures of children's non-cognitive skills, estimated separately for children aged 4–5 and 8–9 attending formal care or education settings. "Negative behaviour" score is reverse-coded.

Figure B.1: Distribution of the interview-based non-cognitive development score at ages 4-5.



Notes: The figure displays the kernel density plot of the interview-based non-cognitive development score for children aged 4-5 attending formal care or education settings.

Table B.2: Measures of teacher and mother perceptions and standardized development measures, by child's age

	Ages 4-5			Ages 8-9		
	N	Mean	SD	N	Mean	SD
Teacher: Cognitive delay	6588	0.15	0.36			
Teacher: Non-cognitive delay	6594	0.21	0.41			
Mother: Non-cognitive delay	7136	0.07	0.25	7772	0.07	0.26
Mother: Cognitive concern	9353	0.08	0.27			
Cognitive score	8664	0.00	1.00	8275	0.00	1.00
WAI score	9069	0.00	1.00			
Non-cognitive score	9180	0.00	1.00	8239	0.00	1.00
Average postcode cognitive score	7381	0.00	1.00	5832	0.00	1.00
Average postcode non-cognitive score	7879	0.00	1.00	5810	0.00	1.00
School contacted about behavior				8284	0.11	0.31

Notes: This table displays summary statistics for teacher and mother perception measures, as well as standardized development measures, by child's age. The sample is limited to children attending formal care or education settings.

B.2 Measures of household control variables

Table B.3: Household control variables, by child's age

	Ages 4-5			Ages 6-7			Ages 8-9		
	N	Mean	SD	N	Mean	SD	N	Mean	SD
Share of Kindergarten cohort	9358	0.53	0.50	8680	0.51	0.50	8397	0.51	0.50
Female child	9358	0.49	0.50	8680	0.49	0.50	8397	0.49	0.50
Age in months	9358	57.25	2.77	8680	81.90	3.23	8397	106.10	3.34
Index of SES	9321	0.00	1.00	8630	0.00	1.00	8329	0.00	1.00
Mother depression	7906	0.00	1.00	8413	0.00	1.00	7717	0.00	1.00
Index: Mothers' interaction with school				3615	0.00	1.00	6948	0.00	1.00

Notes: This table displays summary statistics for household control variables used in the baseline regression analysis by child's age. The sample is limited to children attending formal care or education settings.

B.3 Neighbourhood characteristics

Table B.4 summarizes neighbourhood characteristics computed by the LSAC, based on Census statistics matched to each household's location. The Index of Neighbourhood Relative Advantage and Disadvantage, constructed by the Australian Bureau of Statistics, is a weighted average of indicators related to income, employment, education, and housing (see Australian Bureau of Statistics (2011) for details). I normalize this index by age group. In addition, I control for the age composition of the population and for ethnic and language composition characteristics, as these were not included in the index but may influence the availability of child development services and language development opportunities in the neighbourhood.

Table B.4: Summary of neighbourhood characteristics, by child's age

	Ages 4-5			Ages 6-7			Ages 8-9		
	N	Mean	SD	N	Mean	SD	N	Mean	SD
New South Wales	9358	0.31	0.46	8680	0.31	0.46	8396	0.31	0.46
Share population age 0-4	9358	6.68	1.37	8678	6.54	1.30	8396	5.50	1.67
Share population age 5-9	9358	7.12	1.53	8678	6.87	1.37	8396	4.18	3.18
Index of Neighbourhood Relative Advantage and Disadvantage	9358	0.01	1.00	8678	0.00	1.00	8389	0.00	1.00
Share population Aboriginal	9358	2.19	3.95	8678	2.45	5.12	8396	2.33	5.08
Share population English - first language	9358	87.03	13.04	8678	86.49	13.82	8396	83.93	15.03
Share population born in Australia	9358	86.86	12.41	8678	91.66	8.20	8396	86.56	14.04
Victoria	9358	0.25	0.43	8680	0.24	0.43	8396	0.24	0.43
Queensland	9358	0.21	0.41	8680	0.21	0.41	8396	0.21	0.41
South Australia	9358	0.07	0.25	8680	0.07	0.25	8396	0.07	0.25
Western Australia	9358	0.10	0.30	8680	0.10	0.30	8396	0.10	0.30
Tasmania	9358	0.03	0.16	8680	0.03	0.17	8396	0.03	0.16
Northern Territory	9358	0.01	0.12	8680	0.01	0.11	8396	0.01	0.11
Australian Capital Territory	9358	0.02	0.15	8680	0.02	0.15	8396	0.02	0.16

Notes: This table displays summary statistics for neighbourhood characteristics by child's age. The sample is limited to children attending formal care or education settings.

B.4 Therapy uptake measures

Table B.5 describes therapy uptake for children aged 4–5 and 6–7. The uptake of non-cognitive therapy increases with age, while cognitive therapy uptake declines during the early years of formal schooling.

Table B.5: Summary of therapy uptake measures, by child's age

	Ages 4-5			Ages 6-7		
	N	Mean	SD	N	Mean	SD
Non-cognitive therapy	4374	0.03	0.18	4236	0.07	0.26
Cognitive therapy	4374	0.13	0.33	4236	0.09	0.29

Notes: This table displays summary statistics for therapy uptake measures by child's age. The sample is restricted to children attending formal care or education settings.

B.5 Additional controls

Table B.6 describes the additional controls used in column (3) of Table 2 to account for potential idiosyncratic shocks to children's interview effort at ages 4–5. The sleeping problems score is based on mothers' responses to a Likert-scale question regarding the frequency of their child's sleep difficulties over the past month. Responses are standardized by age group.

During the PPVT test, most parents either remained at a distance or observed the child without intervening. 10 percent actively encouraged their child and 12 percent were not present during the assessment. Most children completed the test without siblings present in the room or with siblings remaining at a distance.

Table B.6: Additional control measures

	Ages 4-5		
	N	Mean	SD
Sibling not present	4319	0.62	0.49
Parent not present	4319	0.12	0.32
Sleeping problems score	7997	0.00	1.00
Parent at a distance	4319	0.42	0.49
Parent observed	4319	0.35	0.48
Parent encouraged	4319	0.10	0.30
Parent interfered	4319	0.01	0.12
Sibling at a distance	4319	0.22	0.42
Sibling observed	4319	0.14	0.35
Sibling encouraged	4319	0.01	0.11
Sibling interfered	4319	0.01	0.09

Notes: This table displays summary statistics for additional control measures for children aged 4–5 attending formal care or education settings. Measures of parent and sibling behaviour during the cognitive test are available only for children in the "Baby" cohort. The sample is restricted to children attending formal care or education settings.

B.6 Childcare arrangement

Table B.7 summarizes early childhood care and education arrangements for children aged 4–5 in the sample. The majority attend preschool or kindergarten, and 26 percent attend daycare. Only a few children in the sample are enrolled in school at this stage.

Table B.7: Childcare arrangements for children aged 4–5

	N	Mean	SD
Child attends daycare	8895	0.26	0.44
Child attends preschool	8895	0.57	0.50
Child attends kindergarten	8895	0.18	0.38
Child attends Grade 1	8895	0.00	0.02

Notes: This table displays summary statistics for childcare arrangements for children aged 4–5 attending formal care or education settings.

Naming conventions and starting ages for early childhood programs vary across Australian states and territories. In the 2000s, *preschool* generally referred to structured, play-based education provided in the year prior to full-time primary school. These programs were delivered in a variety of settings by schools, community organizations, and long-day childcare centers (Australian Bureau of Statistics, 2004). In some jurisdictions, preschool programs are referred to as kindergarten. By contrast, *kindergarten* programs, also known as pre-Year 1, preparatory, reception, transition, or pre-primary, were typically offered in schools during the year before Year 1 and mark the beginning of formal schooling.

During data collection, a substantial number of respondents misreported whether their child attended preschool or kindergarten. This variable was heavily imputed by LSAC using teacher reports and later retrospective reports from caregivers (Australian Institute of Family Studies, 2025). Therefore, in the baseline analysis, I combine these two categories.

Table B.8 reports estimates from the specification with additional controls, as in column (3) of Table 2, distinguishing between preschool and kindergarten attendance. The estimated coefficients on average neighbourhood non-cognitive development are close to those in Table 2, with 0.018 for perceived non-cognitive delays and 0.016 for perceived cognitive delays, with

standard errors essentially unchanged. The latter remains statistically significant at the 5 percent level, while the former falls just outside it.

Table B.8: Teachers' perceptions and average neighbourhood child development

	(1) Non-cognitive delay	(2) Cognitive delay
Neighbourhood	0.018	0.016*
non-cognitive score	(0.009)	(0.007)
Non-cognitive score	-0.032*	-0.028*
	(0.011)	(0.011)
Neighbourhood cognitive	0.001	0.003
score	(0.012)	(0.010)
Cognitive score	-0.074*	-0.087*
	(0.011)	(0.011)
N	1912	1912
R2	0.09	0.13

Notes: Linear probability regression with added controls for sleeping problem intensity, the behaviour of parents and siblings during the test, Australian state/territory dummies, and neighbourhood characteristics from the Census, teacher and classroom characteristics, distinguishing between children attending preschool and kindergarten. Standard errors are clustered at the postcode level. Significance level: * 5 percent.

B.7 Teachers' characteristics

Table B.9 summarizes teachers' characteristics when children are ages 4–5. 63 percent of teachers hold a university degree, and only 16 percent of teachers have five or fewer years of experience. The ages of the youngest and oldest children in the classroom are reported in the teacher questionnaire. On average, the youngest child in the group was 47.5 months old, while the oldest was nearly 65 months old.

Table B.10 examines the extent to which more qualified teachers and higher-quality programs are systematically selected into more developed neighbourhoods. I estimate linear probability models where the dependent variables are teacher and program characteristics, and the independent variables include average neighbourhood cognitive and non-cognitive development, child's age and gender, cohort, household SES index, neighbourhood characteristics, and Australian state/territory dummies.

Table B.9: Teachers' characteristics for children aged 4–5

	Ages 4-5		
	N	Mean	SD
Teacher univ. degree	6327	0.63	0.48
Teaching experience 0-5 years	6329	0.16	0.37
Age of youngest child in class	6188	47.49	10.09
Age of oldest child in class	6181	64.67	8.74
Children to qualified staff ratio	6515	14.39	7.00

Notes: This table displays summary statistics for teacher and program characteristics for children aged 4–5 attending formal care or education settings.

There is no evidence that more qualified teachers systematically select into neighbourhoods with higher average non-cognitive development. This is plausible given that, unlike academic achievement, which is often measured and publicly reported at the community or school level, there is generally no available ranking of programs based on children's non-cognitive skills. Teacher education, which is an important predictor of delay recognition as shown in Subsection 3.3, is not significantly associated with either cognitive or non-cognitive development levels in the neighbourhood.

Table B.11 replicates the analysis in Subsection 3.3, distinguishing between preschool and kindergarten attendance. Among children with low measured development, the coefficient on teacher education remains positive and statistically significant.

Children attending daycare or preschool are less likely to be perceived as having delays than those enrolled in kindergarten programs. Since kindergarten education typically marks the start of formal schooling in Australian states and territories, this suggests that children entering formal education are more likely to be flagged as having developmental delays. This effect appears even among children with high measured development, suggesting that formal school entry does not necessarily improve identification accuracy. One possible explanation is that school teachers may use older children as a reference group, consistent with the relative age effect in teachers' perceptions of delays documented in Elder (2010).

Table B.10: Selection of teachers in more developed neighbourhoods

	Univ. degree	Daycare	Teach. exp. 0-5	Age youngest	Age oldest	Child/staff ratio
Neighbourhood cognitive	-0.012	-0.026*	-0.012	0.418	0.253	-0.295
score	(0.013)	(0.008)	(0.008)	(0.226)	(0.229)	(0.157)
Neighbourhood	0.004	0.001	-0.001	-0.354	0.120	0.064
non-cognitive score	(0.010)	(0.007)	(0.007)	(0.185)	(0.128)	(0.125)
N	5027	6997	5041	4920	4912	5194

Notes: The table reports estimates from separate regressions of the teacher or program characteristic indicated by each column on average neighbourhood non-cognitive and cognitive development. Columns (1)–(3) are linear probability models, while columns (4)–(6) are linear regressions. The sample is limited to children aged 4–5 attending formal care or education settings. Controls include the child's gender, cohort, and age in months; the household socioeconomic status (SES) index; Census neighbourhood characteristics; and Australian state/territory indicators. Standard errors are clustered at the postcode level. Significance level: * 5 percent.

Table B.11: Relationship between teacher and program characteristics and perceived delays in children aged 4–5

	Non-cognitive delay		Cognitive delay	
	(1)	(2)	(3)	(4)
	Non-cogn. score low	Non-cogn. score high	Cogn. score low	Cogn. score high
Teacher univ. degree	0.084*	0.018	0.055*	-0.003
	(0.027)	(0.022)	(0.026)	(0.013)
Teaching experience 0-5	-0.096*	-0.006	0.007	0.032
years	(0.032)	(0.029)	(0.034)	(0.020)
Age of youngest in class	0.002	0.001	0.002	0.001
	(0.002)	(0.001)	(0.002)	(0.001)
Age of oldest in class	0.001	0.000	0.001	0.001
	(0.002)	(0.001)	(0.002)	(0.001)
Children to qualified	-0.002	0.001	0.002	0.001
staff ratio	(0.002)	(0.002)	(0.002)	(0.001)
Child attends daycare	-0.098	-0.111*	-0.204*	-0.084*
	(0.052)	(0.046)	(0.050)	(0.027)
Child attends preschool	-0.052	-0.105*	-0.193*	-0.091*
	(0.043)	(0.039)	(0.045)	(0.023)
N	1364	1407	1213	1484
Mean	0.30	0.16	0.29	0.06
R2	0.07	0.03	0.07	0.04

Notes: Linear probability regression including controls for sleep problem intensity, parent/sibling behavior during testing, Australian state/territory dummies, neighbourhood characteristics from the Census, and teacher/classroom characteristics. The models distinguish between preschool and kindergarten attendance. The sample is limited to children aged 4–5 attending formal care or education settings. Standard errors are clustered at the postcode level. Significance level: * 5 percent.

C Measured development and later outcomes

I examine whether interview-based measures of children's non-cognitive and cognitive development at ages 4–5 predict later outcomes. Table C.1 presents estimates from linear regressions of later child outcomes on interviewer-assessed development measures, controlling for a range of child and household characteristics measured at ages 4–5. The results indicate that higher non-cognitive scores at ages 4–5 are associated with a lower probability of grade repetition by ages 12–13 and with higher scores on Grade 9 reading and numeracy assessments from the National Assessment Program — Literacy and Numeracy (NAPLAN).

Table C.1: Interview development measures at age 4-5 and later child outcomes.

	Repeated grade by ages 12-13	Grade 9 Reading	Grade 9 Math
Non-cognitive score at	-0.010*	4.254*	4.477*
4-5	(0.003)	(0.866)	(0.888)
Cognitive score at 4-5	-0.011*	17.306*	11.831*
	(0.003)	(1.000)	(1.109)
N	6694	6067	5997
Mean	0.05	599.25	607.22

Notes: Linear regressions. Controls include children's gender, cohort, and age in months, household socioeconomic status (SES) index. The sample is limited to children aged 4–5 who attended formal care or education settings. Outcomes are measured at ages 12–13 and in Grade 9. Standard errors are clustered at the postcode level. Significance level: * 5 percent.

D Neighbourhood variation in child development: magnitude and socioeconomic correlates

A maintained assumption throughout the analysis is that the postcode average captures a feature of the local environment that teachers plausibly use as a reference point. This section explores two related questions. First, is there enough between-postcode variation in child development for the average to be meaningful, or is most variation within postcodes? Second, what observable neighbourhood attributes correlate with the between-postcode variation that does exist? This section takes each in turn, first decomposing the total variation into within- and between-

postcode components, and next characterising which neighbourhood attributes correlate with average child development.

If children were distributed across postcodes at random, the postcode average would carry no information about a child beyond sampling noise, and there would be no scope for a reference-group interpretation. To evaluate how much neighbourhood-level variation in skills exists, I decompose the total variation in individual development scores into within- and between-postcode components. Table D.1 reports a one-way analysis of variance of individual interviewer-assessed scores by postcode. The between-postcode share (the intraclass correlation) measures the fraction of total variation in child development that lies across, rather than within, neighbourhoods, and thus the extent to which children are sorted across postcodes on measured development.

Table D.1 reports the decomposition. The intraclass correlation is 0.07 for non-cognitive and 0.11 for cognitive development: most variation in individual development scores lies within postcodes, though a non-negligible share lies between them. Between-postcode variation is larger for cognitive than for non-cognitive development, indicating stronger spatial sorting in cognitive skills. With roughly twenty-six children per postcode, the estimated reliability of the postcode mean is 0.66 for non-cognitive and 0.76 for cognitive development. The postcode averages therefore recover most of the underlying between-postcode signal, and they do so somewhat more precisely for cognitive development.

The final rows of Table D.1 report the correlation between the two postcode averages. This correlation speaks directly to an assumption used in prior work. Lacking an independent measure of non-cognitive skills, one method proposed by Elder and Zhou (2021) assumes that a school's normalized development level is common across skill domains, allowing observed sorting in objective cognitive measures to proxy for sorting in latent non-cognitive skills. The data provide little support for this assumption. The correlation between postcode-average non-cognitive and cognitive development is only 0.215, rising to 0.304 after correcting for attenuation from sampling variation in both averages. Thus, a postcode's relative position in cognitive development is only weakly informative about its position in non-cognitive development, making cognitive development a poor proxy for cross-postcode variation in non-cognitive development.

Next, I estimate the relationship between average child development in a postcode and its

Table D.1: Within- and between-postcode variation in child development

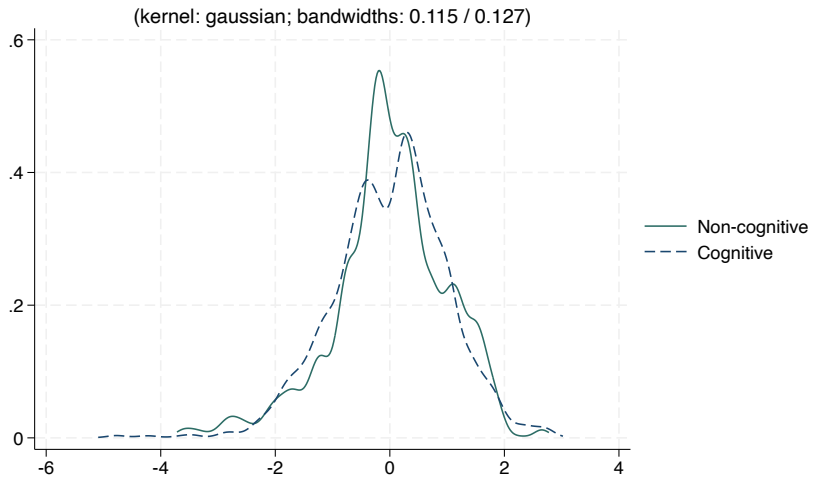
	Non-cognitive development	Cognitive development
Between-postcode SD	0.245	0.326
Within-postcode SD	0.911	0.946
Between-postcode share of variance (ICC)	0.068	0.106
	[0.051, 0.084]	[0.084, 0.128]
Reliability of postcode mean	0.658	0.759
Correlation of postcode means	0.215	
disattenuated	0.304	
Children per postcode	26.5	26.5
Postcodes	278	278
Children	7362	7362

Notes: The table reports a one-way variance decomposition of individual interviewer-assessed development scores by postcode, for children aged 4–5, estimated by analysis of variance. The sample is restricted to postcodes with at least ten other children with non-missing scores. Brackets report asymptotic 95 percent confidence intervals for the intraclass correlation. The number of children per postcode reported is a simple average. The final rows report the Pearson correlation between the postcode-average non-cognitive and cognitive development scores, computed with the postcode as the unit of observation and each postcode weighted equally. The disattenuated correlation divides this figure by the square root of the product of the two reliabilities, and estimates the correlation between the underlying postcode means net of sampling variation in the observed averages.

socioeconomic and demographic composition. The unit of observation is the postcode, and the dependent variable is the average interviewer-assessed development score of children aged 4–5 in the postcode, constructed as in Section 2 and restricted to postcodes with at least ten children with non-missing scores. Figure D.1 shows the distribution of these postcode averages for non-cognitive and cognitive development. I regress each average on the neighbourhood index of relative advantage and disadvantage (standardized), the shares of the population that are Aboriginal, whose first language is English, and who were born in Australia, the shares of the population aged 0–4 and 5–9, and a set of Australian state/territory indicators.

Table D.2 shows that neighbourhood socioeconomic status is a key correlate of average child development. A one standard deviation increase in the index of relative advantage and disadvantage is associated with a 0.07 standard deviation increase in average non-cognitive development and a 0.19 standard deviation increase in average cognitive development. For non-cognitive development the socioeconomic index is the only characteristic significantly related

Figure D.1: Distribution of postcode-average non-cognitive and cognitive development scores, ages 4–5.



Notes: The figure plots kernel densities of the postcode-average non-cognitive and cognitive development scores for children aged 4–5. The sample is restricted to postcodes with at least ten other children with non-missing scores. The unit of observation is the postcode.

to the postcode average. For cognitive development the socioeconomic gradient is nearly three times as steep, and two further characteristics enter significantly: average cognitive development is higher in postcodes with a larger share of residents whose first language is English, and lower in postcodes with a larger share of primary-school-aged children. This pattern is consistent with the nature of the PPVT, which measures receptive English vocabulary and is therefore likely to be particularly sensitive to differences in the local language environment.

Observed neighbourhood characteristics account for a much larger share of the between-postcode variation in average cognitive development than in average non-cognitive development. Together with state fixed effects they explain roughly half of the between-postcode variation in average cognitive development, but only a small fraction of the between-postcode variation in average non-cognitive development. This is consistent with cognitive development being more closely related to observed neighbourhood socioeconomic and demographic conditions, which may partly capture differences in material and educational resources and a stronger spatial sorting in cognitive skills documented above.

Table D.2: Average child development in the neighbourhood and neighbourhood characteristics

	Postcode-average score	
	Non-cognitive	Cognitive
Share aged 0–4	0.003	-0.032
	(0.031)	(0.031)
Share aged 5–9	-0.023	-0.079*
	(0.031)	(0.027)
SEIFA advantage/disadvantage	0.070*	0.189*
	(0.027)	(0.025)
Share Aboriginal	0.005	-0.004
	(0.005)	(0.008)
Share English first language	0.000	0.015*
	(0.004)	(0.005)
Share born in Australia	0.009	0.004
	(0.006)	(0.007)
_cons	-0.559*	-0.919*
	(0.264)	(0.259)
R2	0.08	0.50
State FE	Yes	Yes
N postcodes	278	278

Notes: Each column reports a postcode-level regression with one observation per postcode. The dependent variable is the average interviewer-assessed non-cognitive (column 1) or cognitive (column 2) development score of children aged 4–5 in the postcode, restricted to postcodes with at least ten other children with non-missing scores. Heteroskedasticity-robust standard errors in parentheses. Significance level: * 5 percent.

E Sensitivity to the selected measure of teachers' perceptions

This section estimates how teachers' perceptions of non-cognitive skills vary with the reference group, using an alternative measure of perceptions.

First, I construct a continuous index of perceived non-cognitive problems based on selected items from the Strengths and Difficulties Questionnaire (SDQ), a widely used instrument for assessing children's non-cognitive development (Fiorini and Keane, 2014, Nicoletti and Tonei, 2020). The selected subquestions capture behaviours that align with those evaluated during the interviewer assessment, such as the frequency of temper outbursts, fidgeting, being distracted, and

similar behaviours, as reported by teachers on a Likert scale.

Table E.1 presents results from the regression specified in Equation (1), where the dependent variable is the age-standardized index of non-cognitive problems for children ages 4–5 and 8–9. The coefficient on average neighbourhood non-cognitive development is positive and statistically significant, and is similar in magnitude at both ages.

Table E.1: The relationship between reference group and alternative measure of teacher perceptions of non-cognitive development, by child's age

	Ages 4-5	Ages 8-9
	(1)	(2)
Neighbourhood cognitive	-0.00	-0.00
score	(0.02)	(0.02)
Cognitive score	-0.15*	-0.08*
	(0.02)	(0.01)
Neighbourhood	0.04*	0.03*
non-cognitive score	(0.02)	(0.02)
Non-cognitive score	-0.12*	-0.07*
	(0.02)	(0.02)
N	5174	4841
R2	0.09	0.11

Notes: Linear regression. Controls include children's gender, cohort, and age in months, household socioeconomic status (SES) index. The sample is restricted to children aged 4–5 and 8–9 who attended formal care or education settings. Standard errors are clustered at the postcode level. Significance level: * 5%.

Second, I use the baseline measure of teachers' perceptions to construct an indicator for whether the teacher perceives the child to be advanced relative to children of a similar age. The indicator equals one if the teacher reports that the child is "much more competent than others" in the corresponding developmental domain. This provides a useful test of whether the baseline relationship reflects differences across neighbourhoods in teachers' incentives to identify developmental delays in the presence of resource constraints. Such differences should primarily affect the reporting of delays and need not generate a corresponding relationship for assessments of advanced development. By contrast, if teachers evaluate children relative to local reference points, the same mechanism should operate at both ends of the distribution: conditional on the child's own development, children should be less likely to be perceived as advanced when average development in their neighbourhood is higher. I therefore re-estimate Equation (1), replacing the

perceived-delay outcome with the indicator for advanced development. Table E.2 reports the results.

The estimates are consistent with this prediction for non-cognitive development. Conditional on the child's own development, children living in neighbourhoods with higher average non-cognitive development are significantly less likely to be perceived as advanced relative to children of a similar age. Thus, the neighbourhood relationship is not limited to teachers' identification of developmental delays, as would be expected if it were driven solely by differences in resources or incentives to recognize children requiring additional support.

Table E.2: The relationship between reference group and perceived advanced development.

	Advanced non-cognitive	Advanced cognitive
	(1)	(2)
Neighbourhood cognitive	-0.02*	-0.03*
score	(0.01)	(0.01)
Cognitive score	0.05*	0.08*
	(0.01)	(0.01)
Neighbourhood	-0.02*	-0.01
non-cognitive score	(0.01)	(0.01)
Non-cognitive score	0.01*	0.03*
	(0.01)	(0.01)
N	5251	5247
R2	0.04	0.06

Notes: Linear regression. Control: children's gender, cohort, and age in months, household socioeconomic status (SES) index. The sample is restricted to children aged 4–5 who attended formal care or education settings. Standard errors are clustered at the postcode level. Significance level: * 5%.

F Robustness to measurement error

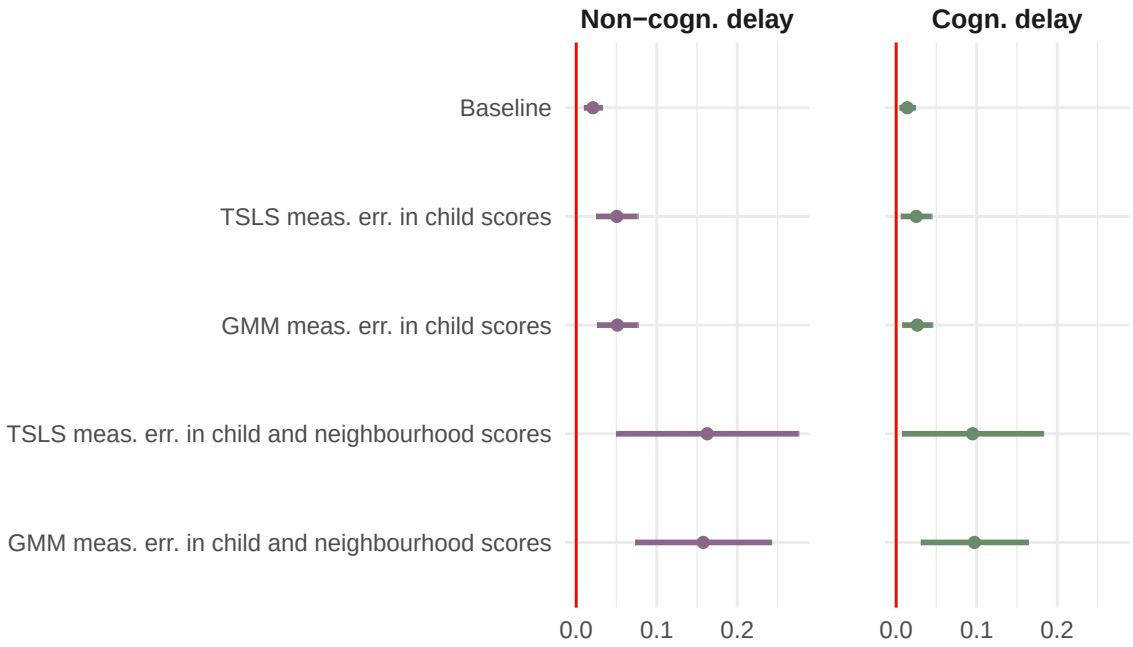
This section examines the sensitivity of the estimates to adjusting for measurement error in both the child's individual development score and the average neighbourhood development score.

Figure F.1 displays estimates of $\beta^{T,N}$ based on Equation (1), using a range of approaches to address measurement error:

- Baseline: Replicates the estimates reported in columns (2) and (4) of Table 1.
- TSLS (measurement error in child scores): Corresponds to column (5) of Table 2. This specification instruments for the child's PPVT score using the WAI score, and for the child's focus during the cognitive test using the degrees of negative and positive behavioural responses.
- GMM (measurement error in child scores): Adjusts for measurement error in children's development scores using GMM estimation. The model jointly estimates the perception equations for cognitive and non-cognitive delays, allowing for correlation in the error terms across equations.
- TSLS (measurement error in child and neighbourhood scores): Extends the previous approach by also instrumenting for measurement error in average neighbourhood development. Specifically, the neighbourhood PPVT average is instrumented with the neighbourhood average WAI score, and neighbourhood focus is instrumented with average negative and positive behavioural responses.
- GMM (measurement error in child and neighbourhood scores): Uses GMM to adjust for measurement error in both the child and neighbourhood scores, estimating the cognitive and non-cognitive perception equations jointly and allowing for correlated errors across outcomes.

Taken together, the estimates reported in Figure F.1 imply that accounting for measurement error increases the magnitude of the estimated association between average neighbourhood non-cognitive development ($\beta^{T,N}$) and teachers' perceptions, although the estimates become noisier.

Figure F.1: Estimated coefficient on the average neighbourhood non-cognitive development - adjustment for classical measurement error



Notes: This figure shows the estimated coefficient on average neighbourhood non-cognitive development along with 95 percent confidence intervals from regressions of perceived delays on child and neighbourhood cognitive and non-cognitive development levels, adjusting for measurement error in child or (and) neighbourhood development scores. All models control for child gender, cohort, age in months, and household socioeconomic status (SES) index. The sample is restricted to children aged 4–5 who attended formal care or education settings. Standard errors are clustered at the postcode level. Significance level: * 5 percent.

G Teachers' perceptions and qualifications: split-sample estimates

Table 4 lets the child and neighbourhood development coefficients differ by teacher qualifications through interaction terms, holding the remaining controls and the residual variance common across teachers. This appendix reports a complementary specification that estimates the baseline teacher equation separately for subsamples of children split by their teacher's qualification group, so that every coefficient, and the residual variance, is free to differ across groups. For each developmental domain I estimate Equation 1 on two subsamples defined by teacher education (a university degree versus a diploma or certificate) and, separately, by teacher experience (five or fewer years versus six or more). Each regression includes the same teacher and classroom characteristics as Table 4, with the characteristic that defines the split omitted within subsamples, where it is constant.

I test the equality of coefficients for the child's own development and average neighbourhood development levels across subsamples by seemingly unrelated estimation, which stacks the two group equations and forms a joint cluster-robust covariance matrix. The resulting p-values, reported in the lower panel of each table, test whether the own-development alignment and reference-group coefficients differ between subsamples.

Tables G.1 and G.2 report the estimates, and the patterns are consistent with the interacted specification. For teacher education, the equality test rejects for the coefficient on the child's own cognitive score for perceptions of both non-cognitive and cognitive delays. Therefore, university-trained teachers' perceptions track measured cognitive development more closely. At the same time, the tests for the two neighbourhood coefficients do not reject, providing no evidence that the reference-group gradient differs by teacher education.

For teacher experience, the differences are concentrated in perceptions of non-cognitive delays, where the equality test is rejected for the child's own non-cognitive score and for the average neighbourhood non-cognitive score. The corresponding tests for perceptions of cognitive delays are not rejected. Less-experienced teachers' perceptions of non-cognitive delays are thus less closely aligned with children's measured non-cognitive development and more strongly asso-

ciated with the local non-cognitive environment, with no corresponding difference for perceived cognitive delays.

Table G.1: Teachers' perceptions and education: split-sample estimates

	Non-cognitive delay		Cognitive delay	
	(1) Diploma/cert.	(2) Univ. degree	(3) Diploma/cert.	(4) Univ. degree
Non-cognitive score	-0.035*	-0.054*	-0.032*	-0.034*
	(0.011)	(0.008)	(0.009)	(0.007)
Cognitive score	-0.030*	-0.068*	-0.060*	-0.098*
	(0.010)	(0.008)	(0.009)	(0.007)
Neighbourhood non-cognitive score	0.016	0.028*	0.021*	0.016*
	(0.009)	(0.008)	(0.008)	(0.007)
Neighbourhood cognitive score	0.002	0.008	0.009	0.019*
	(0.010)	(0.008)	(0.009)	(0.007)
N	1721	2843	1718	2843
R2	0.06	0.08	0.08	0.12
Equality of coefficient across degree groups (p-value):				
Non-cognitive score	0.210		0.884	
Cognitive score	0.011		0.002	
Neighbourhood non-cognitive score	0.351		0.633	
Neighbourhood cognitive score	0.690		0.367	

Notes: The table reports split-sample estimates of the baseline linear probability model for perceived delays, estimated separately for teachers holding a university degree and teachers holding a diploma or certificate. Columns (1)–(2) report perceived non-cognitive delays and columns (3)–(4) perceived cognitive delays. Controls include the child's gender, cohort, and age in months, a household socioeconomic status (SES) index, an indicator for daycare attendance (versus kindergarten or preschool), the age range of children in the class, the child-to-qualified-staff ratio, and teacher experience. Teacher education, which defines the split, is constant within subsamples and therefore omitted. The lower panel reports p-values from tests of the equality of each coefficient across the two subsamples, computed by seemingly unrelated estimation with a joint covariance matrix clustered at the postcode level. Significance level: * 5 percent.

Table G.2: Teachers' perceptions and experience: split-sample estimates

	Non-cognitive delay		Cognitive delay	
	(1)	(2)	(3)	(4)
	6+ years	0–5 years	6+ years	0–5 years
Non-cognitive score	-0.053*	-0.016	-0.036*	-0.025
	(0.007)	(0.016)	(0.006)	(0.015)
Cognitive score	-0.054*	-0.060*	-0.088*	-0.070*
	(0.007)	(0.016)	(0.006)	(0.015)
Neighbourhood non-cognitive score	0.018*	0.047*	0.016*	0.025*
	(0.007)	(0.014)	(0.005)	(0.013)
Neighbourhood cognitive score	0.003	0.023	0.014*	0.019
	(0.007)	(0.016)	(0.006)	(0.014)
N	3865	699	3862	699
R2	0.07	0.08	0.11	0.08
Equality of coefficient across experience groups (p-value):				
Non-cognitive score	0.031		0.553	
Cognitive score	0.683		0.263	
Neighbourhood non-cognitive score	0.022		0.462	
Neighbourhood cognitive score	0.171		0.768	

Notes: The table reports split-sample estimates of the baseline linear probability model for perceived delays, estimated separately for teachers with six or more years of experience (6+ years) and teachers with five or fewer years (0–5 years). Columns (1)–(2) report perceived non-cognitive delays and columns (3)–(4) perceived cognitive delays. Controls include the child's gender, cohort, and age in months, a household socioeconomic status (SES) index, an indicator for daycare attendance (versus kindergarten or preschool), the age range of children in the class, the child-to-qualified-staff ratio, and teacher education. Teacher experience, which defines the split, is constant within subsamples and therefore omitted. The lower panel reports p-values from tests of the equality of each coefficient across the two subsamples, computed by seemingly unrelated estimation with a joint covariance matrix clustered at the postcode level. Significance level: * 5 percent.

H Teachers' and mothers' perceptions – classical measurement error adjusted

This section examines whether the relationships between average neighbourhood child development, teachers' perceptions and mothers' perceptions described in Equation (3) and Table 5 are sensitive to classical measurement error in children's measured development. As in the measurement-error exercise for teachers' perceptions, I replace the baseline non-cognitive development index with the child's degree of focus during the cognitive assessment and estimate a two-stage least squares specification in which the individual non-cognitive and cognitive development measures are instrumented with alternative measures of development. For each age group, I also estimate the corresponding OLS specification using the same development measures and the same estimation sample. Average neighbourhood child development is not significantly associated with mothers' perceptions, consistent with the baseline results.

Table H.1: The relationship between mothers' perceptions of non-cognitive delay, average neighbourhood child development, and teacher perceptions

	Ages 4–5		Ages 8–9	
	(1) OLS	(2) ME adj.	(3) OLS	(4) ME adj.
Neighbourhood non-cognitive score	-0.008	0.002	-0.000	0.020
	(0.005)	(0.011)	(0.003)	(0.013)
Neighbourhood cognitive score	0.002	-0.006	-0.005	-0.004
	(0.008)	(0.012)	(0.004)	(0.007)
Teacher: Non-cognitive delay	0.079*	0.058		
	(0.023)	(0.030)		
Teacher: Cognitive delay	0.023	-0.014		
	(0.025)	(0.038)		
School contacted about behavior			0.110*	0.090*
			(0.016)	(0.020)
Non-cognitive score (focus item)	-0.012	-0.294	-0.007	-0.160
	(0.006)	(0.161)	(0.004)	(0.090)
Cognitive score	-0.000	0.018	-0.003	-0.022
	(0.008)	(0.062)	(0.004)	(0.019)
Mother depression score	0.016*	0.025*	0.013*	0.014*
	(0.008)	(0.011)	(0.005)	(0.005)
Mother-school interaction index			0.001	0.004
			(0.004)	(0.004)
Lag mother: Non-cognitive delay	0.347*	0.355*	0.516*	0.513*
	(0.049)	(0.056)	(0.031)	(0.032)
N	1613	1613	3916	3916
Mean	0.06	0.06	0.07	0.07

Notes: The table reports estimates from linear probability regressions in which the dependent variable indicates that the mother perceives a non-cognitive delay. The sample is restricted to children who attend formal care or education settings, with their mother as the respondent to the face-to-face interview. For each age group, the OLS and measurement-error-adjusted specifications are estimated on the same sample. The adjusted specifications treat the child's focus measure and cognitive score as measured with error and use the positive- and negative-behaviour ratings and an alternative cognitive assessment as instruments. At ages 4–5, the alternative cognitive assessment is the WAI score, at ages 8–9, it is the MR score. Lagged maternal perceptions are measured at ages 2–3 and 6–7, respectively. Controls include children's gender, cohort, and age in months, the household socioeconomic status (SES) index, the mother's depression score, Census neighbourhood characteristics, and Australian territory dummies. The ages 8–9 specifications additionally control for the mother's school involvement. Standard errors are clustered at the postcode level. Significance level: * 5 percent.

I Heterogeneity of therapy uptake association with perceptions by mother's education

This section analyzes the heterogeneity in the relationship between therapy uptake and perceptions, as described in Section 5, across households with more- and less-educated mothers. I estimate the linear value-added regressions outlined in Equation (4) separately for two subsamples of children: those whose mother holds a university degree and those whose mother does not. Rather than controlling for a composite SES index, I include controls for the mother's age, number of children, family income, marital status, whether English is the household language, and employment status. All regressions additionally control for lagged dependent variables and children's cognitive and non-cognitive development scores. The "OLS" columns include average neighbourhood development scores and neighbourhood characteristics from the Census, and the "Postcode fixed-effects" columns absorb these neighbourhood terms. The "Interacted" column includes interactions between the perception measures and an indicator for whether the mother has a degree, and is estimated on the full sample.

Tables I.2 and I.1 show no strong evidence of heterogeneity in the role of teachers' perceptions by maternal education. Non-cognitive delays perceived by mothers remain the strongest predictor of non-cognitive therapy uptake across both education groups. Mothers' concerns about cognitive delays are a somewhat stronger predictor of cognitive therapy uptake among mothers with a university degree, although column (5) shows that the interaction between maternal degree status and cognitive concerns is not statistically significant.

Table I.1: Delay recognition by teachers and mothers and cognitive therapy, heterogeneity by mother's education

	No degree		Univ. degree		Interacted
	(1) OLS	(2) Postcode FE	(3) OLS	(4) Postcode FE	(5) Postcode FE
Neighbourhood non-cognitive score at 4–5	0.018*		0.012		
	(0.006)		(0.007)		
Neighbourhood cognitive score at 4–5	0.010		-0.001		
	(0.007)		(0.009)		
Teach.: Non-cognitive delay at 4–5	0.015	0.014	-0.006	-0.007	0.015
	(0.016)	(0.023)	(0.021)	(0.033)	(0.022)
Teach.: Cognitive delay at 4–5	0.078*	0.085*	0.043	0.007	0.087*
	(0.018)	(0.030)	(0.028)	(0.055)	(0.028)
Moth.: Non-cognitive delay at 4–5	0.058*	0.072	0.013	0.034	0.057
	(0.025)	(0.042)	(0.029)	(0.041)	(0.038)
Moth.: Concern cognitive delay at 4–5	0.025	0.042	0.157*	0.181*	0.043
	(0.023)	(0.036)	(0.034)	(0.070)	(0.035)
Non-cognitive score at 4–5	-0.011	-0.013	-0.009	-0.005	-0.011
	(0.006)	(0.009)	(0.009)	(0.014)	(0.007)
Cognitive score at 4–5	-0.029*	-0.031*	-0.005	-0.008	-0.021*
	(0.007)	(0.010)	(0.009)	(0.012)	(0.007)
Therapy at 4–5	0.202*	0.187*	0.245*	0.257*	0.217*
	(0.017)	(0.029)	(0.022)	(0.052)	(0.024)
Teach.: Non-cognitive delay × Univ. degree					-0.001
					(0.036)
Teach.: Cognitive delay × Univ. degree					-0.067
					(0.055)
Moth.: Non-cognitive delay × Univ. degree					-0.042
					(0.054)
Moth.: Concern cognitive delay × Univ. degree					0.113
					(0.067)
N	2140	2561	1214	1462	4023
Mean	0.08	0.08	0.07	0.08	0.08

Notes: Linear regression using the sample of children aged 6–7 who attend formal care or education settings, with their mother as the respondent to the face-to-face interview. Columns (1)–(2) restrict to children whose mother does not hold a university degree; columns (3)–(4) to those whose mother does. Lagged variables are measured when children are ages 4–5. Controls: lag dependent variable, lag cognitive and non-cognitive score, children's gender, cohort, age in months, mother's age, number of siblings, family income, mother's marital status, whether English is the household language, and mother's employment status. OLS columns additionally control for neighbourhood-average development scores and neighbourhood characteristics from the Census; the postcode fixed-effects columns absorb all neighbourhood terms. Standard errors are clustered at the postcode level. Significance level: * 5%.

Table I.2: Delay recognition by teachers and mothers and non-cognitive therapy, heterogeneity by mother's education

	No degree		Univ. degree		Interacted
	(1) OLS	(2) Postcode FE	(3) OLS	(4) Postcode FE	(5) Postcode FE
Neighbourhood non-cognitive score at 4–5	0.009		-0.003		
	(0.005)		(0.006)		
Neighbourhood cognitive score at 4–5	-0.004		0.010		
	(0.006)		(0.008)		
Teach.: Non-cognitive delay at 4–5	0.010	-0.004	0.032	0.030	0.005
	(0.013)	(0.019)	(0.018)	(0.026)	(0.018)
Teach.: Cognitive delay at 4–5	0.014	0.035	0.002	-0.018	0.035
	(0.015)	(0.023)	(0.024)	(0.046)	(0.022)
Moth.: Non-cognitive delay at 4–5	0.183*	0.162*	0.137*	0.143*	0.159*
	(0.020)	(0.045)	(0.025)	(0.064)	(0.043)
Moth.: Concern cognitive delay at 4–5	0.018	0.024	0.066*	0.115	0.029
	(0.018)	(0.027)	(0.029)	(0.071)	(0.027)
Non-cognitive score at 4–5	0.002	0.001	-0.005	-0.007	-0.001
	(0.005)	(0.007)	(0.008)	(0.012)	(0.005)
Cognitive score at 4–5	0.004	0.001	-0.008	-0.002	0.001
	(0.005)	(0.007)	(0.007)	(0.010)	(0.005)
Therapy at 4–5	0.101*	0.094	0.038	0.081	0.098*
	(0.026)	(0.054)	(0.034)	(0.080)	(0.040)
Teach.: Non-cognitive delay					0.023
× Univ. degree					(0.030)
Teach.: Cognitive delay					-0.051
× Univ. degree					(0.043)
Moth.: Non-cognitive delay					-0.033
× Univ. degree					(0.070)
Moth.: Concern cognitive delay					0.073
× Univ. degree					(0.066)
N	2140	2561	1214	1462	4023
Mean	0.05	0.06	0.05	0.05	0.06

Notes: Linear regression using the sample of children aged 6–7 who attend formal care or education settings, with their mother as the respondent to the face-to-face interview. Columns (1)–(2) restrict to children whose mother does not hold a university degree; columns (3)–(4) to those whose mother does. Lagged variables are measured when children are ages 4–5. Controls: lag dependent variable, lag cognitive and non-cognitive score, children's gender, cohort, age in months, mother's age, number of siblings, family income, mother's marital status, whether English is the household language, and mother's employment status. OLS columns additionally control for neighbourhood-average development scores and neighbourhood characteristics from the Census; the postcode fixed-effects columns absorb all neighbourhood terms. Standard errors are clustered at the postcode level. Significance level: * 5%.

J Mothers' perceptions and home environment

In addition to the uptake of therapy services, mothers who believe their children are falling behind may engage in different parental investment strategies. In this section, I examine the relationship between mothers' perceptions of non-cognitive delays at ages 8–9 and various aspects of the home environment measured at ages 10–11. These include the use of community parenting resources, parenting style, frequency of development-promoting activities (such as reading to or with the child), parental expectations about child's future, and tutoring.

Table J.1 summarizes home environment measures when children are aged 8–9 and 10–11. These include parenting practices, use of community parenting resources and professional services, household members' weekly educational activities with the child, total time spent by mothers and fathers with the child, and mothers' expectations about the child's future educational attainment.

Community resource uptake includes the use of parenting education programs and support groups. Mothers are asked whether anyone in the family has used parenting education courses or programs, or contacted a parent support group or helpline within the past 12 months. Around 4 percent of mothers report using such resources when children are aged 8–9 and 10–11.

Parenting style measures include maternal warmth and maternal anger. Both scores are constructed by LSAC and standardized by age. The maternal warmth score is the average of self-reported responses to a battery of items measuring warm, affectionate behaviour. The anger score averages responses to items reflecting disapproval, anger, and lack of praise in interactions with the child.

The measure of tutoring or additional help frequency is based on mothers' responses to whether the child received any tutoring or help from someone outside the household in the past 12 months. I construct an indicator equal to one if the mother reports any use of tutoring during this period. On average, 15 percent of households report using additional help for children aged 8–9.

Mothers' educational aspirations are captured from the question: "Looking ahead, how far do you think the study child will go in his/her education?" I construct an indicator equal to one

if the mother expects the child to complete a university degree or postgraduate qualification. Around 67 percent of mothers expect their child to earn a college degree.

Several additional variables capture the frequency with which household members engage in educational activities with the child. For activities reported over the past week, such as reading, I transform categorical responses into weekly frequencies as follows: 0 for"Not in the past week," 1.5 for "1 or 2 days," 4 for "3–5 days," and 6.5 for "6–7 days." On average, household members read to the child slightly more than 2 times per week when children are aged 8–9, declining to slightly more than 1 time per week at ages 10–11.[44]

Finally, mothers report spending an average of 26 hours per week in childcare when children are aged 8–9, decreasing to less than 22 hours at ages 10–11. Fathers report a lower time investment, averaging less than 12 hours per week.[45]

[44]Talking about school is elicited on a different scale. Mothers report how often they and the study child talk about the child's school activities, and I convert responses to weekly frequencies as 7 for "Daily," 3.5 for "A few times a week," 1 for "About once a week," 0.5 for "A few times a month," and 0 for "Rarely or never."

[45]The exact question asked was: "How much time per week do you spend actively doing things with your children (for example, playing with them, helping with personal care, teaching, coaching or actively supervising them, getting them to childcare, school or other activities)?"

Table J.1: Summary of home environment measures, by child's age

	Ages 8-9			Ages 10-11		
	N	Mean	SD	N	Mean	SD
Used parenting education courses or programs last 12 months	8360	0.04	0.20	7857	0.04	0.20
Used parent support groups or info from phone/net last 12 months	8360	0.04	0.20	7857	0.04	0.19
Use tutor	4059	0.15	0.36	7656	0.16	0.37
Mother warmth score	7671	0.00	1.00	7635	0.00	1.00
Mother anger score	7668	0.00	1.00	7635	0.00	1.00
Mother expects child univ. degree	8079	0.67	0.47	3560	0.67	0.47
Weekly times talk school	8349	6.60	1.18	7869	6.61	1.17
Weekly times read	8358	2.18	2.33	4160	1.11	1.84
Mother average weekly time with child	8195	26.15	17.67	3637	21.81	15.32
Father average weekly time with child	5159	11.97	9.78	2334	11.39	9.46

Notes: This table displays summary statistics for home environment measures by child's age. The sample is restricted to children attending formal care or education settings.

To assess how family investment decisions vary with maternal perceptions, I estimate a linear regression in which the dependent variable represents various measures of family-based investment at ages 10–11 ($I^F_{i,t}$), and the key independent variable is the lagged indicator of mothers' perceptions of non-cognitive developmental delays at ages 8–9 ($M_{i,t-1}$):

$$I^F_{i,t} = \beta^{F,M} M_{i,t-1} + \beta^{F,X}_t X^F_{i,t} + \beta^{F,I} I^F_{i,t-1} + \epsilon^F_{i,t}. \tag{5}$$

Following the estimation approach described in Section 5, I address potential endogeneity in parental perceptions by using lagged maternal perceptions as an independent variable to mitigate reverse causality and adopting a value-added approach with a range of controls for potential confounding factors. $X^F_{i,t}$ is a vector of controls that includes the child's gender, cohort, and age in months, as well as the household SES index. Control variables also include the mother's depressive symptoms score, neighbourhood characteristics, Australian state/territory dummies, lagged non-cognitive and cognitive scores, and lagged investment $I^F_{i,t-1}$. The standard errors are clustered at the postcode level.

Table J.2 presents estimates of Equation (5) for various home environment measures when children are aged 10–11. Delays perceived by mothers at ages 8–9 predict two noticeable types of parental responses. Panel A shows that, on the one hand, mothers who perceive delays are more likely to engage in compensatory investments. They are more likely to use parenting education courses and programs, participate in support groups and helplines, and hire a tutor or extra help for their child, similar to the findings of Kinsler and Pavan (2021). Combined with the increased use of therapy services, this implies that parents who perceive developmental delays are more likely to seek professional support for their children.

On the other hand, Panel B shows that mothers who perceive non-cognitive delays also report different attitudes toward their children. They exhibit lower levels of warmth and higher levels of anger in their parenting practices. They also report lower educational expectations: those who believe their children have a non-cognitive delay are 7 percentage points less likely to expect that their children will obtain a university education.

Interventions aimed at alleviating parental misperceptions about child development have been found to be effective in improving parental investment choices and even children's out-

Table J.2: Delays perceived by mothers and family investments for children ages 10–11

	Mother: child more difficult at 8-9				
Dependent variable at 10-11	Coef.	SE	N	Mean	R^2
A. Uptake of professional services					
Used parenting education courses or programs	0.05*	(0.01)	6615	0.04	0.04
Used parent support groups or helplines	0.05*	(0.01)	6615	0.04	0.06
Used tutor	0.10*	(0.03)	3314	0.16	0.15
B. Parental attitudes and quality of interactions					
Mother Warmth Score [SD]	-0.15*	(0.04)	6615	0.01	0.47
Mother Anger Score [SD]	0.28*	(0.04)	6613	-0.01	0.44
Mother expects child univ. degree	-0.07*	(0.03)	3078	0.68	0.41
C. Weekly quality time investment					
Read to child	0.36*	(0.14)	3282	1.13	0.16
Talk about school	-0.05	(0.06)	6572	6.66	0.11
Mom childcare time	-0.67	(0.97)	3327	21.96	0.11
Dad childcare time	-0.16	(0.75)	1756	10.76	0.20

Notes: The table reports estimates from a linear regression using a sample of children aged 10–11 who attend formal care or education settings, with their mother as the respondent to the face-to-face interview. Lagged variables are measured at ages 8–9. Controls include children's gender, cohort, and age in months; the household socioeconomic status (SES) index; neighbourhood characteristics from the Census; Australian state/territory dummy; mother's depression score; lag of non-cognitive and cognitive scores; and lag of the dependent variable. Standard errors are clustered at the postcode level. Significance level: * 5 percent.

comes, at least in the short run (Dizon-Ross, 2019; Bergman, 2021). My results imply that while parents might respond to updated perceptions about non-cognitive delay in children by increasing compensatory investments like therapy and tutoring, they may also experience a decrease in the quality of parent-child interactions which have been shown to play a key role in children's non-cognitive development (Fiorini and Keane, 2014; Falk et al., 2021). Therefore, informing parents about non-cognitive delays in their children may need to be accompanied by additional parenting support to mitigate potential negative effects on parental attitudes.